

AFBR-57D7APZ

850nm Digital Diagnostic SFP+ Transceiver
for 8.5/4.25/2.125 Gigabit Fibre Channel



Characterization Report

Description

The AFBR-57D7APZ optical transceivers are reduced cost SFP+ 8.5-4.25-2.125 Gigabit Fibre Channel modules for use on multimode fiber.

The specifications for this transceiver are from the FC-P1-4 revision 8.00 standard and SFF-8431 rev 3.0 and INF-8074i MSA's.

The following transceiver parameters are characterized over voltage and temperature:

- Supply Current (ICC)
- Power Dissipation (PWR)
- Transmitter Light Output Power (TX LOP)
- Transmitter Optical Modulation Amplitude (TX OMA)
- Extinction Ratio (ER)
- Transmitter Optical Mask Margin (MM)
- Transmitter Optical Mask Margin with Worst-case Input Jitter
- Transmitter Optical Rise/Fall Times (TX TR and TX TF)
- Transmitter Waveform Distortion Penalty (TWDP)
- Transmitter Optical Center Wavelength
- Transmitter Optical Spectral Width
- Transmitter RIN12OMA
- Transmitter Encircled Flux
- Transmitter Electrical S-Parameters (SDD11 and SCD11)
- Receiver Sensitivity (SENS)
- Stressed Receiver Sensitivity (RX SRS)
- Receiver Eye Mask Margin (RX MM)
- Receiver Differential Output Voltage Modulation Amplitude (RX VMA)
- Receiver Output Total Jitter (RX TJ)
- Receiver Output Deterministic Jitter (RX DJ)
- Receiver Output Data Dependent Pulse Width Shrinkage (RX DDPWS)
- Receiver Electrical S-Parameters (SDD22 and SCC22)
- Receiver Common Mode Output Voltage (CMOV)
- Receiver Reflectance
- Receiver Maximum Input Power
- Loss of Signal Assert Level (LOSA)
- Loss of Signal De-assert Level (LOSD)
- Loss of Signal Hysteresis (LOSH)
- DMI Voltage Accuracy (DMI VCC)
- DMI Temperature Accuracy (DMI TEMP)
- DMI Laser Output Power Accuracy (DMI LOP)
- DMI RX Input Power Accuracy (DMI RX)
- Inrush Current During Hotplug
- LOS and TX DISABLE Timing
- Module Initialization Timing
- Power Supply Noise Rejection (PSNR)
- Live Traffic ESD
- EMI
- Radiated Immunity

Definition of Terms

Various standards in the optical networking industry provide definitions for these terms and recommended test and measurement methods. Applicable references include FC-P1-4 and IEEE 802.3.

Independent Variables

Supply voltage The DC voltage measured at the SFP connector pin. Voltage is varied from 2.97 to 3.63V.

Temperature The case temperature in degrees C as measured with a thermocouple placed on the back of the module behind the optical connector port receptacle. Temperature is varied from -15C to 90C for most tests.

Dependent Variables

Supply Current (ICC) The supply current is defined as the total DC current in mA delivered to the combination of the VCCR and VCCT pins of the SFP+.

Power Dissipation (PWR) The power consumption in mW is the total power utilized by the module.

Transmitter Average Optical Power (TX LOP) The transmitter average launch power is defined as the average optical power in dBm at the output of the optical fiber.

Transmitter Optical Modulation Amplitude (TX OMA) The transmitter OMA is defined as the difference in uW between a logic-1 and a logic-0 at the output of the optical fiber.

Transmitter Extinction Ratio (ER) The transmitter extinction ratio is defined as $10 \cdot \log_{10}(PL1/PL0)$ in dB, where PL1 is the optical power in the logic-1 state and PL0 is the optical power in the logic-0 state. This measurement is informative.

Transmitter Optical Mask Margin (TX MM) The transmitter mask margin is measured through a fourth-order Bessel-Thomson filter as defined in FC-PI-4 rev. 8.00.

Transmitter Optical Mask Margin with Worst Case Input Jitter The transmitter mask margin is measured through a fourth-order Bessel-Thomson filter with worst-case input jitter as defined in FC-PI-4 rev. 8.00.

Transmitter Rise/Fall Times (TX TR and TX TF) The transmitter rise and fall times are defined as the 20%-80% time in ps. This measurement is informative.

Transmitter Waveform Distortion Penalty (TWDP) The transmitter waveform distortion penalty is a measure of the deterministic penalty of the waveform from a particular transmitter and reference emulated multimode fibers, with a reference receiver.

Transmitter Uncorrelated Jitter (TX UJ) The transmitter uncorrelated jitter is a measure of any jitter that is not correlated to the data stream.

Transmitter Center Wavelength The transmitter center wavelength is the value of the central wavelength of the operating, modulated laser.

Transmitter Spectral Width The transmitter spectral width is the weighted root mean square width of the optical spectrum.

Transmitter RIN OMA The transmitter RIN OMA describes the contributions of laser intensity fluctuations to the electrical noise in the receiver relative to the signal power. RIN OMA is measured using a high-speed optical plug-in with a filtered bandwidth of 10.3125 GHz. This measurement is informative.

Transmitter Encircled Flux The encircled flux is the optical power distribution relative to the center of the optical fiber core.

Transmitter Electrical S-Parameters (SDD11 and SCD11) The differential return loss and the differential to common mode conversion looking from the host system into the module

Receiver Sensitivity (RX SENS) The receiver sensitivity is the minimum detectable optical modulation amplitude in dBm OMA of a compliant optical waveform at which a Bit Error Rate of 10^{-12} is achieved.

Receiver Optical Stressed Receive Sensitivity (RX SRS) The stressed receive sensitivity is the minimum optical power in OMA required to achieve a Bit Error Rate 10^{-12} with a degraded RX input eye. At 8.5 Gbps, the degraded RX input eye must have a vertical eye closure of 3.5 dB at and jitter components as specified by FC-PI-4.

Receiver Differential Eye Mask Margin with Stressed RX Input (RX MM) The differential output eye mask margin is measures the margin to the output mask of FC-PI-4. The RX input is the same degraded input used in the stressed receive sensitivity measurement.

Receiver Differential Output Voltage Modulation Amplitude (RX VMA) The differential output voltage modulation amplitude is the total peak-to-peak voltage in mV from the module's receive channel.

Receiver Total Jitter (RX TJ) Total jitter measured when a stressed optical input source is present.

Receiver Deterministic Jitter (RX DJ) Deterministic jitter measured when a stressed optical input source is present.

Receiver Data Dependent Pulse Width Shrinkage (RX DDPWS) Data dependent pulse width shrinkage measured when a stressed optical input source is present.

Receiver Electrical S-Parameters (SDD22, SCC22 and SCD22) The differential and common mode return loss and the differential-to-common mode conversion looking into the receiver outputs from the host system.

Receiver Common Mode Output Voltage (CMOV) The common mode output voltage is an RMS measurement, calculated by applying the histogram function over one UI to the common mode signal. The common mode signal is the average of the (+) and (-) outputs from the receiver.

Receiver Reflectance Receiver reflectance is the ratio in dB of reflected to incident optical power to the receiver.

Receiver Maximum Input Power The maximum input power that can be applied to the module without bit errors.

Loss of Signal Assert Level (LOSA) The loss of signal assert level is the optical power level in dBm OMA that causes the LOS output pin to switch from "0" to "1".

Loss of Signal De-assert Level (LOSD) The loss of signal de-assert level is the optical input power in dBm OMA that cause the LOS output pin to switch from "1" to "0".

Loss of Signal Hysteresis (LOSH) The loss of signal hysteresis is the difference in dB between the LOS de-assert and assert levels.

Diagnostic Monitoring Interface Voltage Accuracy (DMI VCC) Diagnostic monitoring interface voltage accuracy is the difference in V between the supply voltage reported by the module and the actual measured supply voltage at the input to the module.

Diagnostic Monitoring Interface Temperature Accuracy (DMI TEMP) Diagnostic monitoring interface temperature accuracy is the difference in degrees C between the temperature reported by the module and the actual case temperature of the module as measured at the location specified by the Avago datasheet.

Diagnostic Monitoring Interface Voltage Accuracy (DMI LOP) Diagnostic monitoring interface laser output power accuracy is the difference in dB between the laser output power reported by the module and the actual measured laser output power.

Diagnostic Monitoring Interface Voltage Accuracy (DMI RX) Diagnostic monitoring interface receiver input power accuracy is the difference in dB between the receiver input power reported by the module and the actual receiver input power. The worst case error from measurements at -3, -6, -9 and -12 dBm input LOP is reported.

Inrush Current During Hotplug The inrush current during hotplug is specified by the slope of the input current and maximum current drawn by the module.

LOS and TX_DISABLE Timing The LOS timing describes the time required for the module to assert or de-assert the LOS signal when the receiver input power crosses the threshold. The TX_DISABLE timing describes the time required for the module to bring the laser to 90% of its nominal output power when the TX_DISABLE is de-asserted and the time required for the laser to turn off when the TX_DISABLE is asserted.

Module Initialization Timing The module initialization time is the time from initial power until the point when the module is ready to pass data and respond to A0 and A2 inquiries.

Power Supply Noise Rejection (PSNR) DPSNR performance is the optical mask margin penalty and the receiver sensitivity penalty when power supply noise is applied per SFF-8431 rev 3.0. A 100 mV max signal was applied; SFF-8431 rev 3.0 specifies 66 mV.

Live Traffic ESD +/-15 kV air and +/-8 kV contact ESD pulses are applied to various points on the module and the fiber cable connector per IEC-61000-4-2 and while traffic is being passed. One pulse per second is applied. Bit errors are counted. If less than 1% of the packets per second have errors, then the module passes. The input power to the receiver is the unstressed sensitivity level.

EMI Electromagnetic interference generated by the module. Measured per FCC Part 15 and CISPR 22 standards.

Radiated Immunity Sensitivity penalty due to incident EM fields, per IEC 61000-4-3.

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Table 1. Test Conditions for Selected Parameters

Transmitter						
Test Parameter	Unit	VCC	Temperature	Rate (Gbps)	Pattern	Details
Average Optical Power	dBm	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	FC-PI-4
Optical Modulation Amplitude	uW	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	111110000	
Extinction Ratio	dB	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	Informative
Mask Margin	%	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5, 4.25, 2.125	PRBS-7	FC-PI-4
Electrical S-Parameters	dB	3.3V	25°C	n/a	n/a	SFF-8431 rev 3.0
Receiver						
Test Parameter	Unit	VCC	Temperature	Rate (Gbps)	Pattern	Details
Sensitivity	dBm OMA	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5, 4.25, 2.125	PRBS-7	FC-PI-4 (Informative)
Stressed Receiver Sensitivity	dBm OMA	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	JSPAT	FC-PI-4
Receiver Output Jitter	UI	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	JSPAT	FC-PI-4
Electrical S-Parameters	dB	3.3V	25°C	n/a	n/a	FC-PI-4
Maximum Input Power	dBm	3.3V +/- 5%	-15°C, 25°C, 90°C	8.5	PRBS-7	FC-PI-4
LOS Assert and De-assert	dBm OMA	3.3V +/- 5%	-15°C, 25°C, 90°C	8.5	PRBS-7	
LOS Hysteresis	dB	3.3V +/- 5%	-15°C, 25°C, 90°C	8.5	PRBS-7	
DMI						
Test Parameter	Unit	VCC	Temperature	Rate (Gbps)	Pattern	Details
Voltage	V	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	
Temperature	°C	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	
Laser Output Power	dBm	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	
Receiver Power	dBm	3.3V +/- 10%	-15°C, 25°C, 90°C	8.5	PRBS-7	
Sample size is 25 pieces unless otherwise noted in the report.						

Test Results

Supply Current (ICC)

Table 2. ICC (mA), 210 mA max

	ICC								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	129	140	145	137	147	153	148	159	165
Min	126	134	138	135	145	149	146	155	161
Max	132	144	150	142	150	156	151	161	167

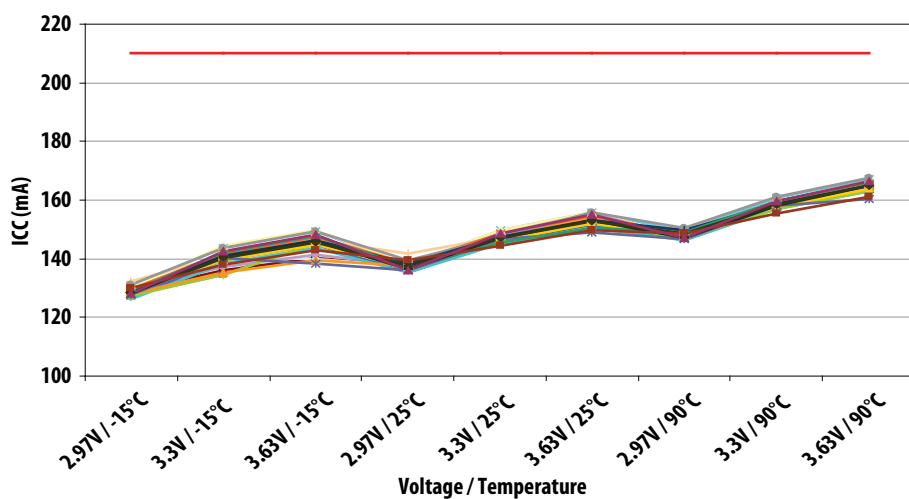


Figure 1. ICC over the 9 test conditions.

Power Dissipation (PWR)

Table 3. PWR (mW), 765 mW max

	PWR								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	382	462	527	408	486	555	440	523	597
Min	375	444	202	402	477	540	435	513	583
Max	393	476	544	421	494	565	447	532	608

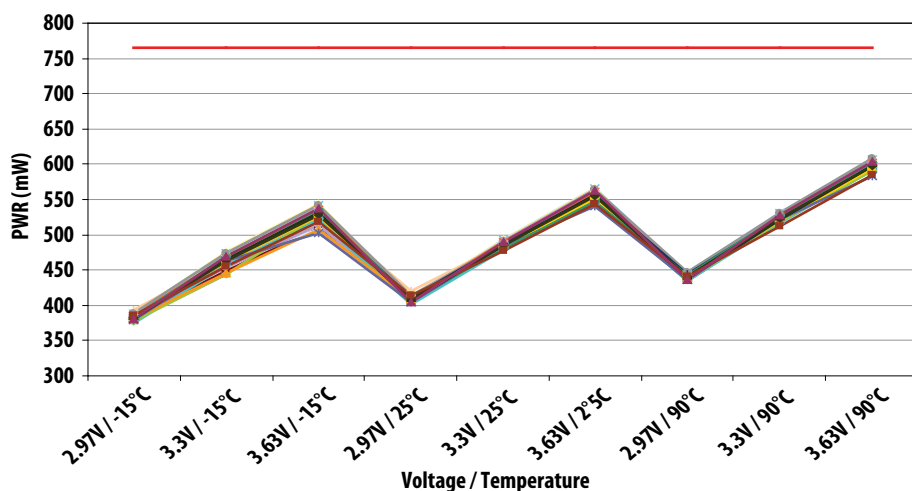


Figure 2. PWR over the 9 test conditions.

Average Optical Power (TX LOP)

Table 4. TX LOP (dBm), -8.2 dBm min

	TX LOP								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-2.42	-2.44	-2.46	-2.55	-2.56	-2.57	-3.29	-3.23	-3.16
Min	-2.64	-2.66	-2.68	-2.82	-2.81	-2.83	-3.75	-3.67	-3.60
Max	-2.09	-2.15	-2.14	-2.28	-2.18	-2.18	-2.96	-3.87	-3.88

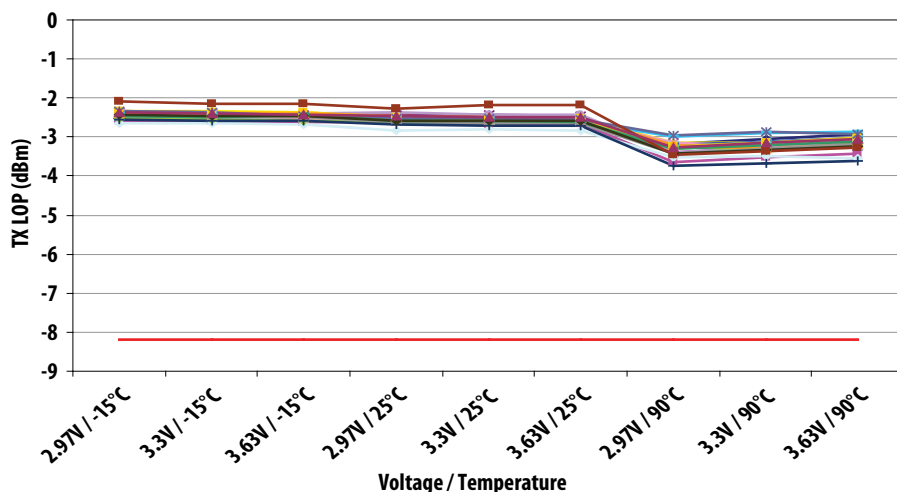


Figure 3. LOP over the 6 test conditions.

Transmitter Optical Modulation Amplitude

Table 5. TX OMA (uW), 302 uW min

	TX OMA								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	416	443	478	475	511	562	379	406	445
Min	365	382	417	424	445	498	317	345	385
Max	480	509	561	539	575	626	434	460	511

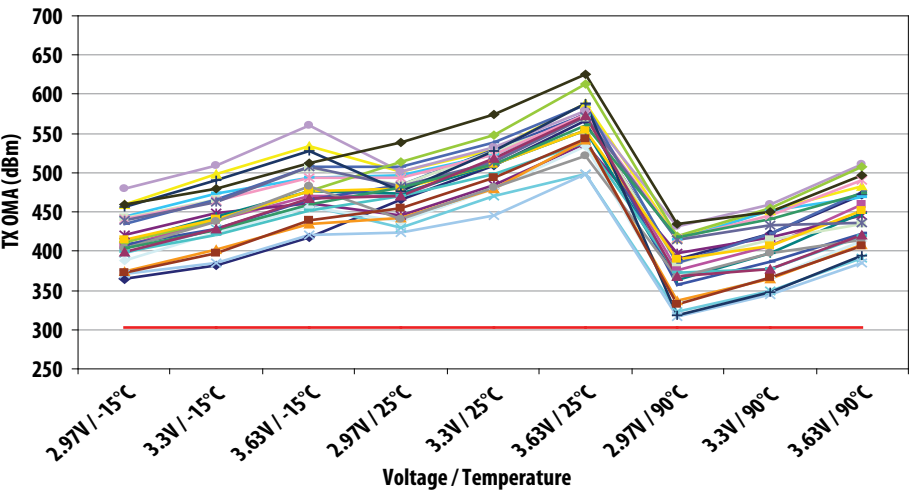


Figure 4. TX OMA over the 9 test conditions.

Transmitter Extinction Ratio

Table 6. ER (uW), informative

	ER								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	3.08	4.07	3.85	3.66	4.28	4.81	3.60	4.26	4.73
Min	2.77	3.07	2.62	3.46	3.45	4.39	3.28	3.70	4.04
Max	3.89	5.09	5.53	4.20	4.93	5054	4.22	4.78	5.59

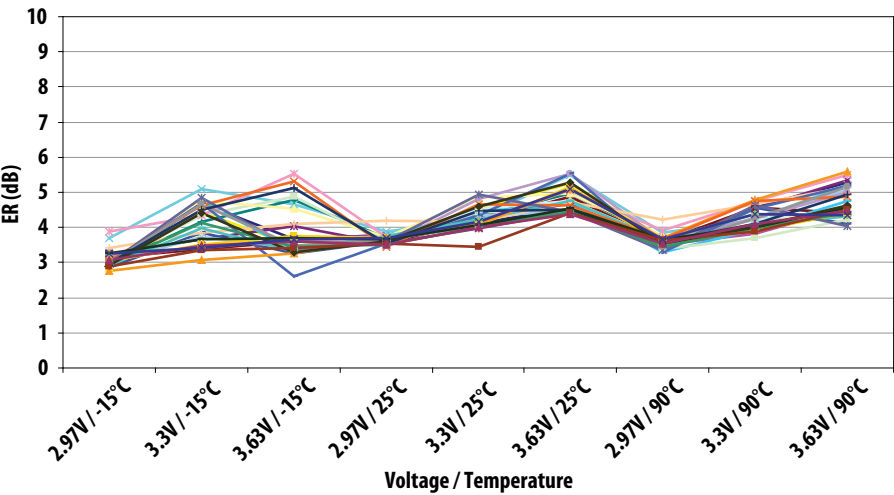


Figure 5. ER over the 9 test conditions.

Transmitter Optical Mask Margin (MM)

Table 7. 8.5 Gbps MM (%), 0% min

	8.5 Gbps MM								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	41	37	36	44	42	39	39	38	35
Min	34	26	23	38	37	32	33	30	25
Max	48	46	44	48	49	47	44	43	42

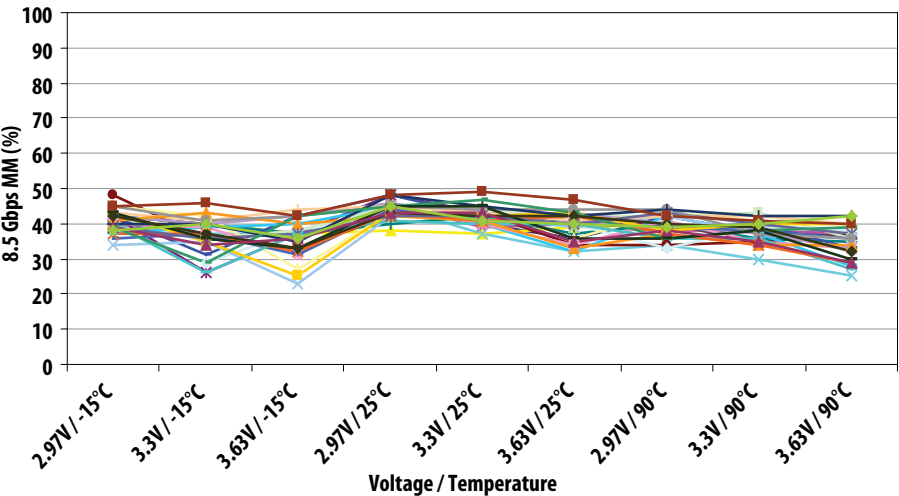


Figure 6. 8.5 Gbps MM over the 9 test conditions.

Table 8. 4.25 Gbps MM (%), 0% min

	4.25 Gbps MM								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	40	40	41	42	43	43	40	41	41
Min	33	29	32	33	33	32	33	32	33
Max	54	53	53	53	50	49	52	51	51

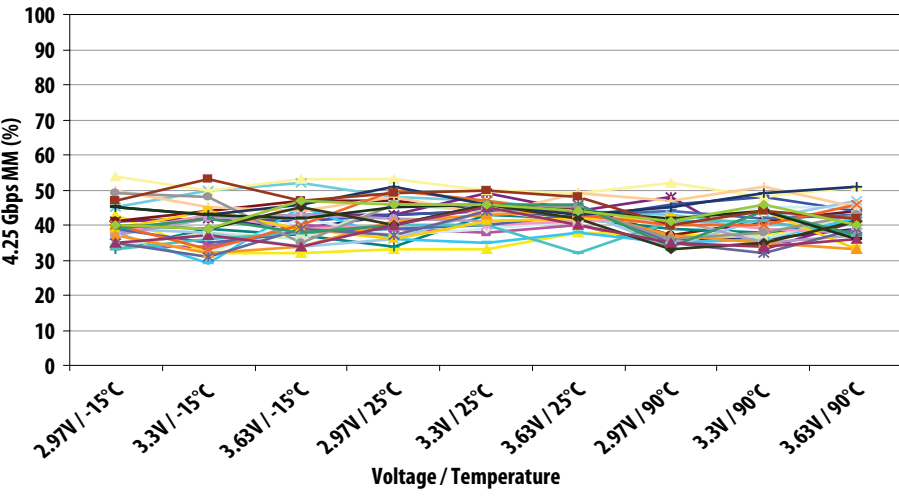


Figure 7. 4.25 Gbps MM over the 9 test conditions.

Transmitter Optical Mask Margin (MM)

Table 9. 2.125 Gbps MM (%), 0% min

	2.125 Gbps MM								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	58	61	60	60	62	63	55	59	60
Min	51	54	51	54	57	56	43	51	54
Max	69	68	66	70	69	69	64	64	66

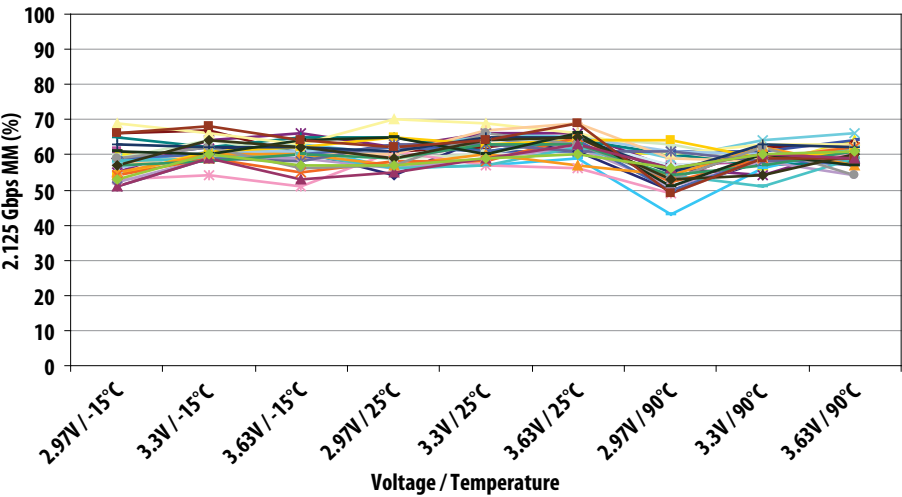


Figure 8. 2.125 Gbps MM over the 9 test conditions.

Transmitter Optical Mask Margin with Worst Case Input Jitter

Table 10. MM with Worst Case Input Jitter (%), 0% min, Sample Size 5

	MM with Worst Case Input Jitter				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	16	19	18	16	14
Min	14	16	16	12	12
Max	20	20	21	18	18

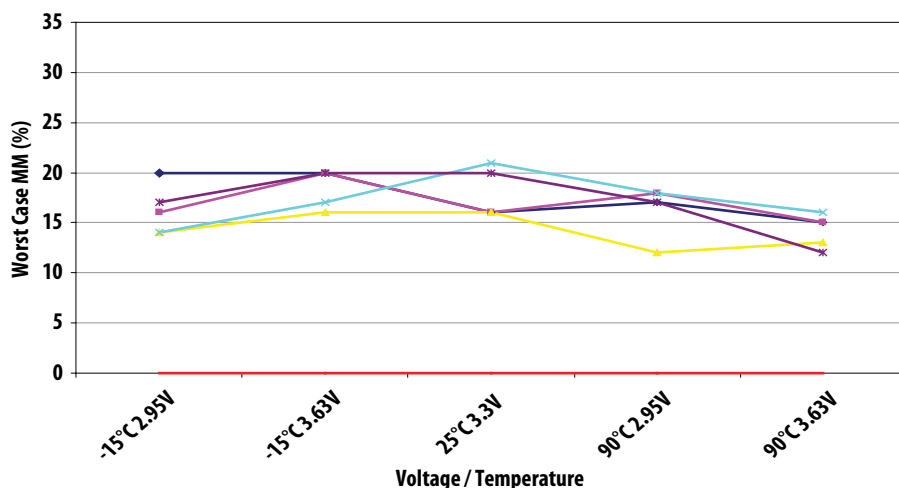


Figure 9. Transmitter optical mask margin with worst case input jitter over the 5 test conditions.

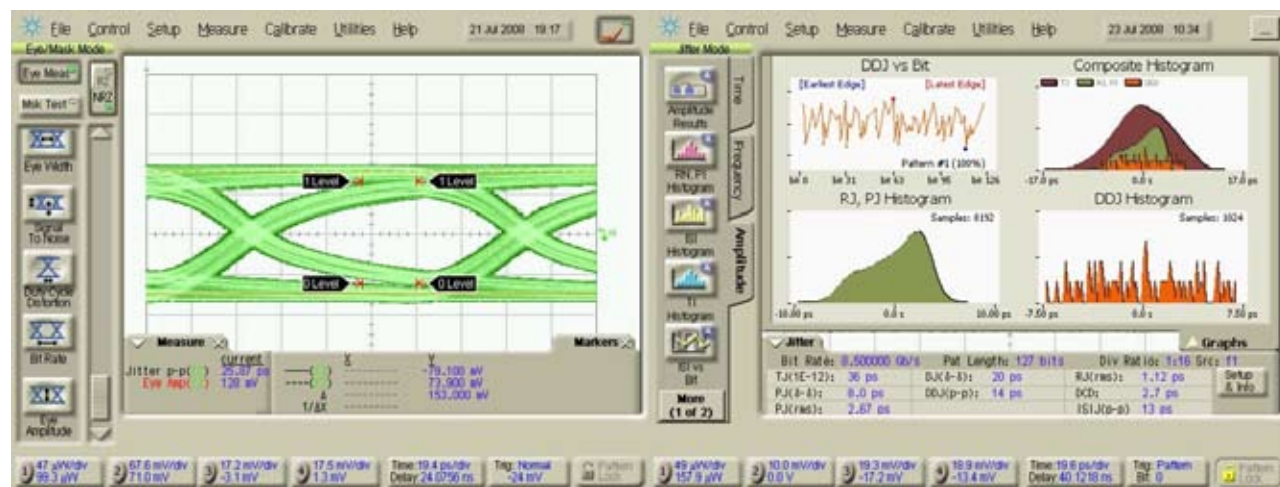


Figure 10. Electrical input signal for worst case mask margin measurements.

Transmitter Rise Time (TX TR)

Table 11. TX TR (ps), informative

	TX TR								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	36.2	35.0	35.1	38.8	38.1	37.6	38.3	37.5	37.5
Min	33.5	30.5	30.1	35.1	33.4	31.8	34.7	32.7	32.7
Max	38.8	37.8	38.2	41.6	40.8	40.7	42.2	41.2	42.1

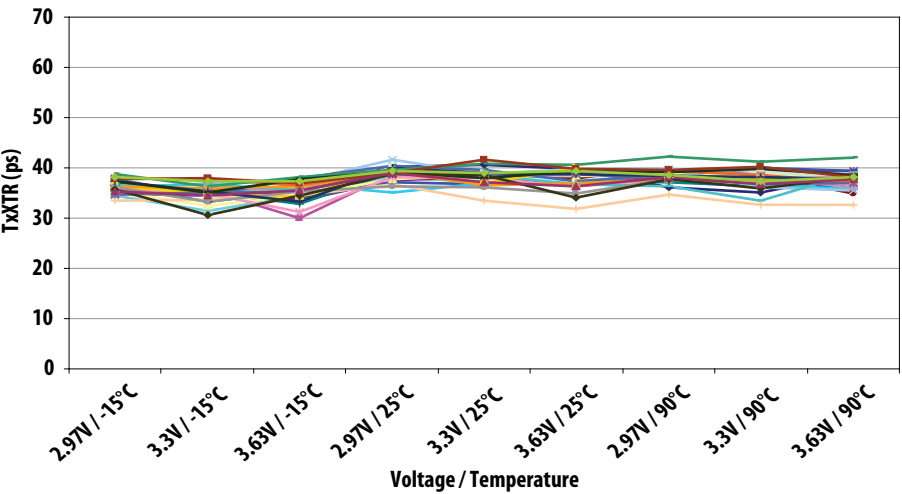


Figure 11. TX TR over the 9 test conditions.

Transmitter Fall Time (TX TF)

Table 12. TX TF (ps), informative

	TX TF								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	41.3	42.7	43.0	42.0	43.0	43.9	47.9	48.9	49.5
Min	38.8	40.4	40.8	40.3	40.7	42.0	46.1	47.0	47.4
Max	43.0	46.4	47.3	44.2	45.4	46.2	51.7	51.8	52.4

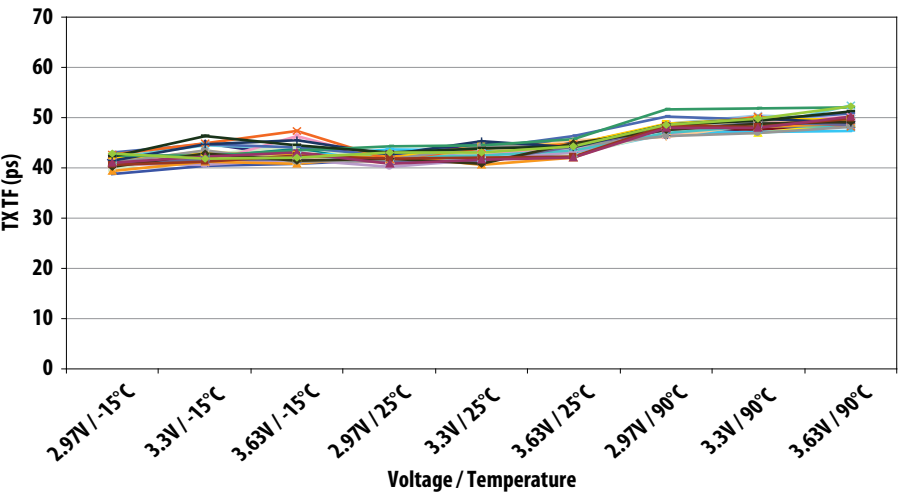


Figure 12. TX TF over the 9 test conditions.

Transmitter Waveform Distortion Penalty (TWDP)

Table 13. TWDP (dB), 4.3 dB max

	TWDP				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	3.93	3.94	4.00	3.97	3.96
Min	3.84	3.85	3.91	3.86	3.89
Max	4.02	4.01	4.09	4.06	4.08

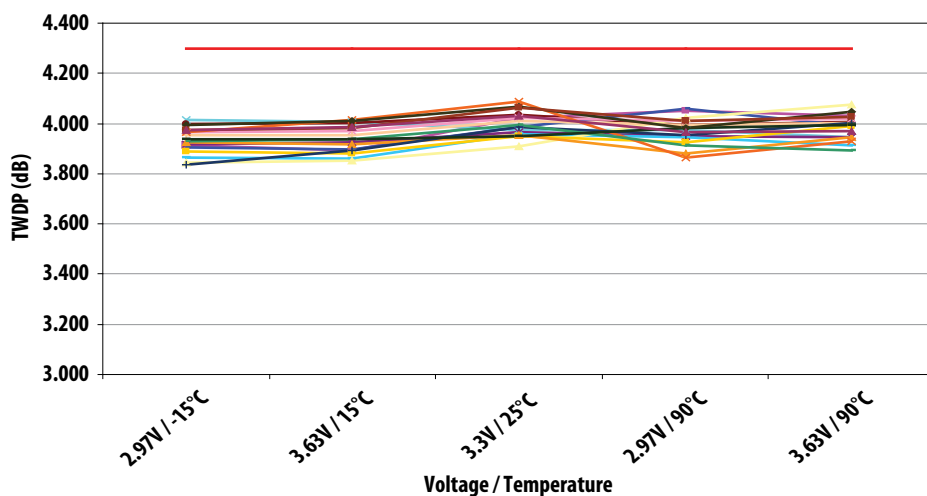


Figure 13. TWDP over the 5 test conditions.

Transmitter Uncorrelated Jitter (TX UJ)

Table 14. TX UJ (UI rms), 0.03 UI max

	TX UJ				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	0.013	0.012	0.012	0.013	0.013
Min	0.011	0.011	0.010	0.012	0.011
Max	0.017	0.015	0.017	0.016	0.016

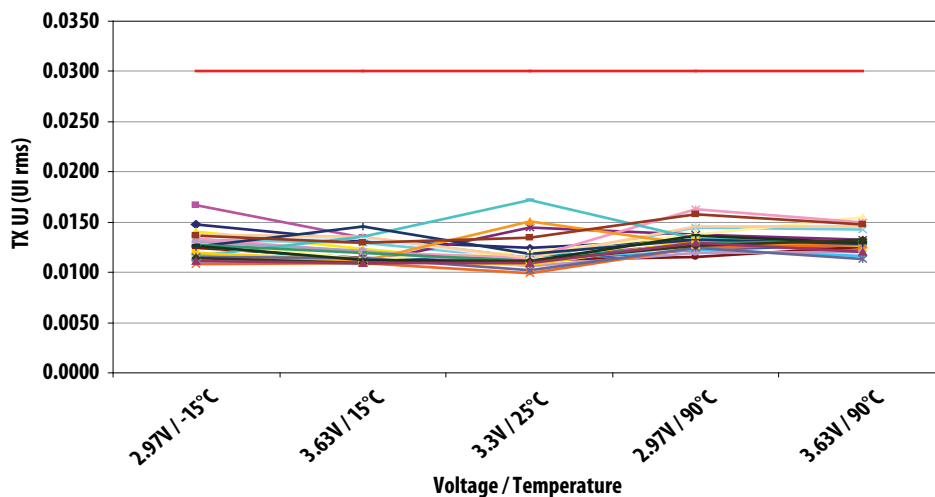


Figure 14. TX UJ over the 5 test conditions.

Center Wavelength

Table 15. Center Wavelength (nm), 840-860 nm

	Center Wavelength				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	842.7	842.7	845.2	849.5	849.6
Min	842.1	842.1	844.7	849.1	849.2
Max	843.4	843.5	845.8	850.0	850.1

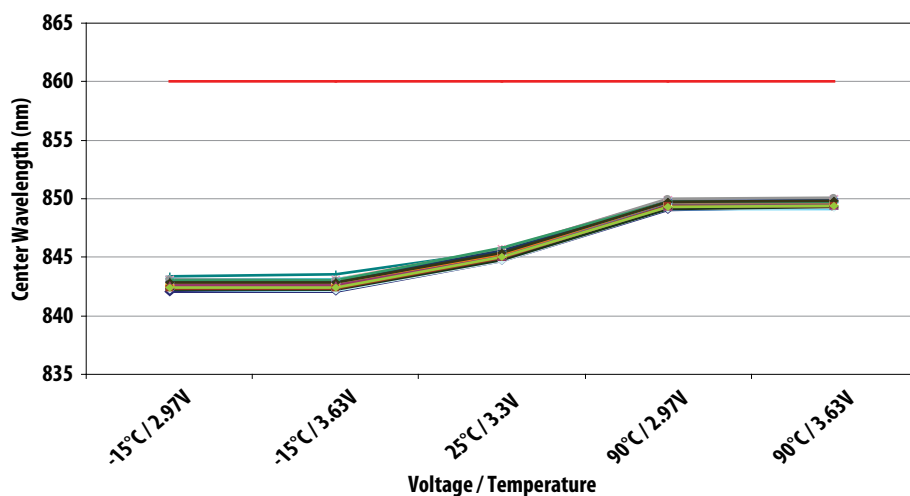


Figure 15. Center wavelength over the 5 test conditions.

Spectral Width

Table 16. Spectral Width (nm), 0.65 nm max

	LOS Assert Timing				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	0.318	0.315	0.275	0.235	0.237
Min	0.263	0.254	0.238	0.200	0.202
Max	0.386	0.381	0.352	0.271	0.274

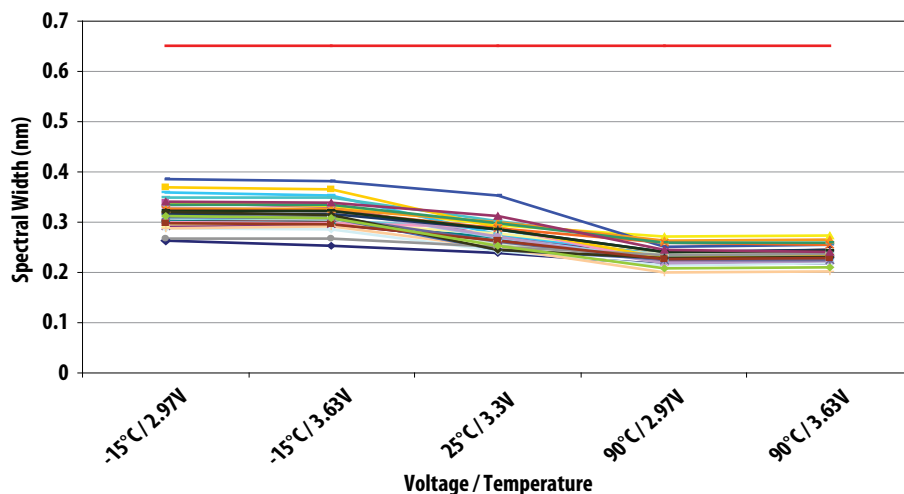


Figure 16. Spectral width over the 5 test conditions.

RIN OMA

Table 17. RIN OMA (dB/Hz), -128 dB/Hz max

	RIN OMA				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	-131.4	-131.6	-133.3	-132.9	-133.3
Min	-132.9	-133.1	-135.2	-134.2	-135.1
Max	-129.8	-130.2	-131.2	-130.2	-130.9

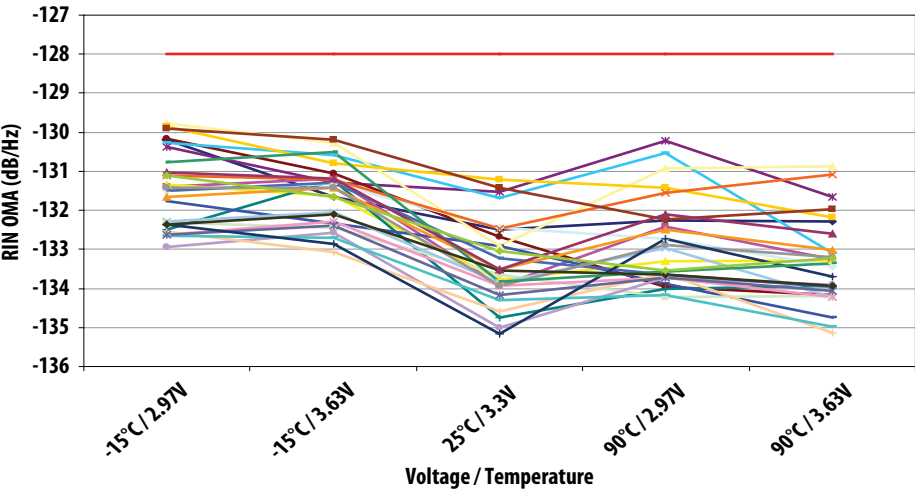


Figure 17. RIN OMA over the 5 test conditions.

Transmitter Encircled Flux

Per standard

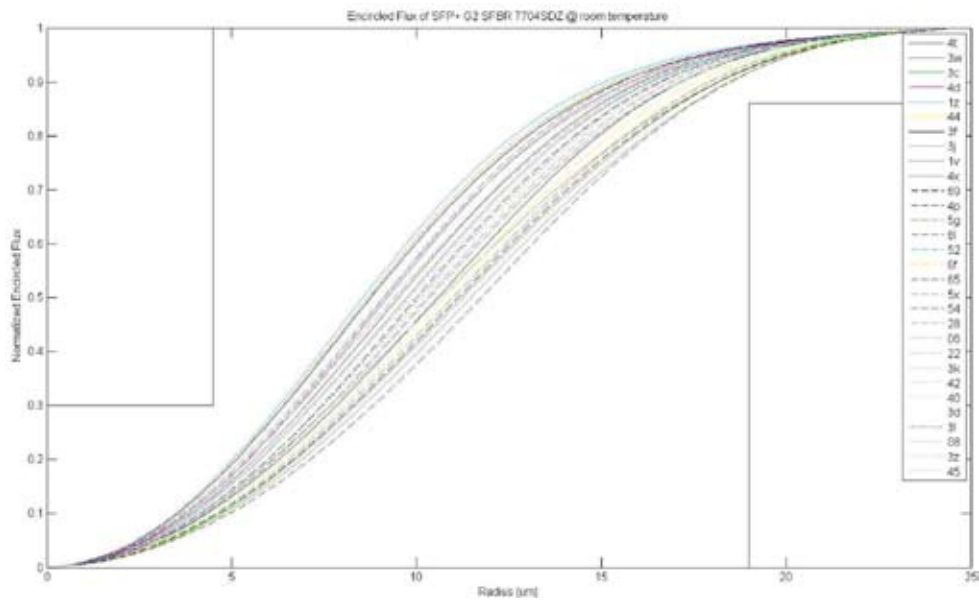


Figure 18. Encircled flux at 25°C.

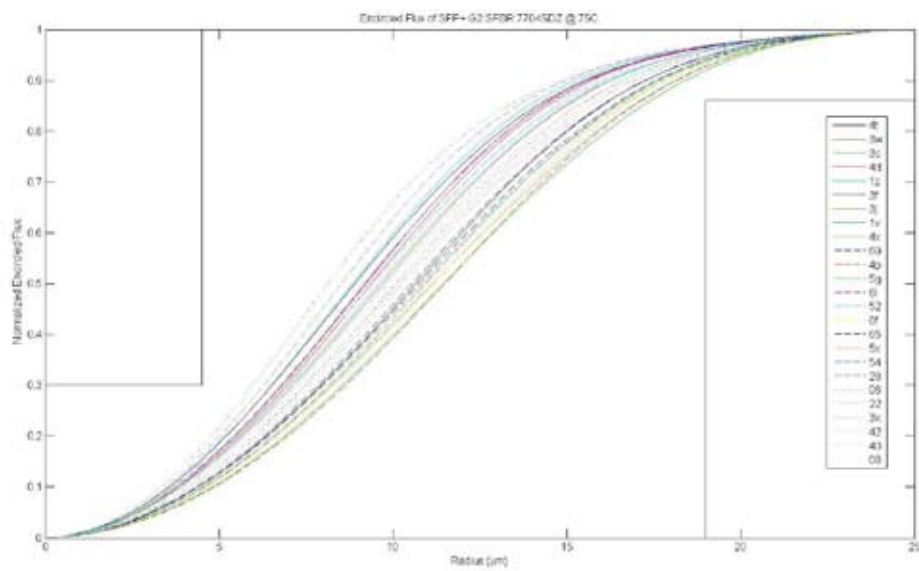


Figure 19. Encircled Flux at 75°C.

Transmitter S-Parameters (SDD11 and SCD11)

Spec per FC-PI-4 rev 8.0. 25/25 modules pass. Typical data are shown.

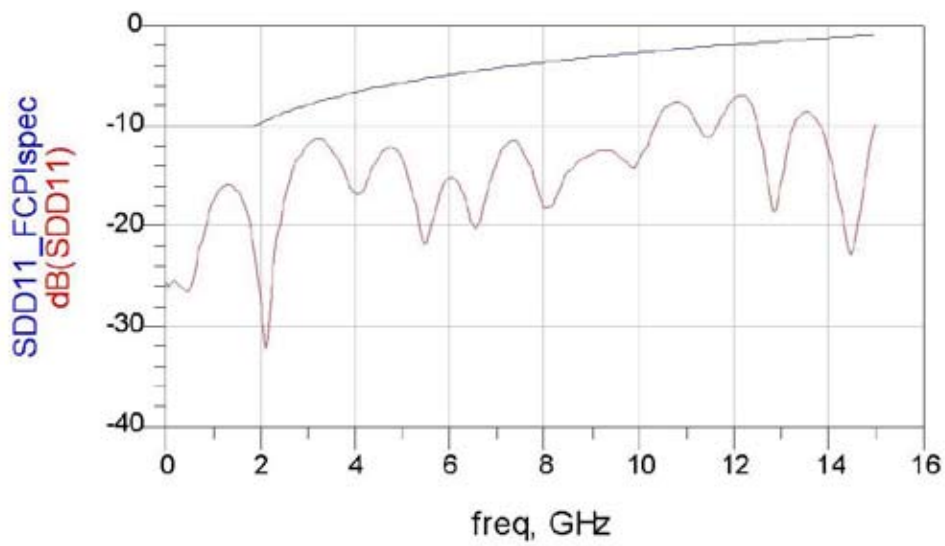


Figure 20. SDD11 at 25°C / 3.3V.

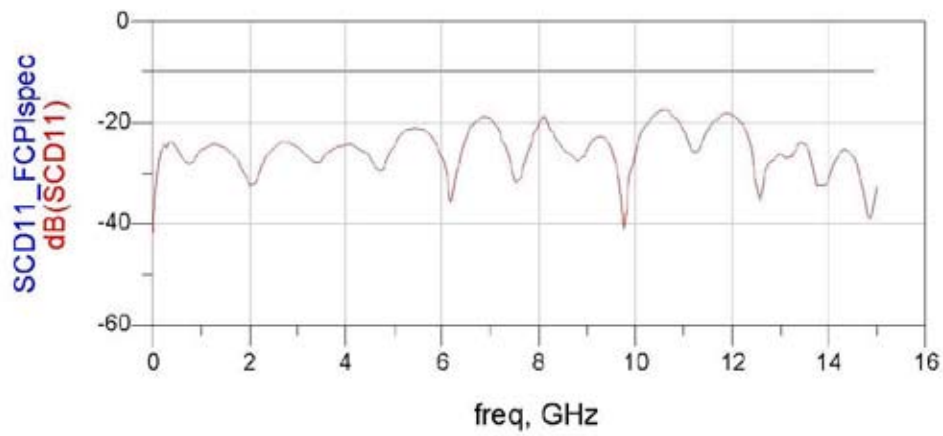


Figure 21. SCD11 at 25°C / 3.3V.

Receiver Sensitivity (SENS)

Table 18. 8.5 Gbps SENS (dBm OMA), -11.2 dBm OMA max

	8.5 Gbps SENS								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-17.0	-16.7	-16.7	-16.8	-16.6	-16.5	-16.4	-16.4	-16.3
Min	-17.7	-17.3	-17.5	-17.4	-17.3	-17.3	-17.0	-16.9	-17.1
Max	-16.0	-15.8	-15.8	-15.8	-15.6	-15.5	-15.8	-15.8	-15.4

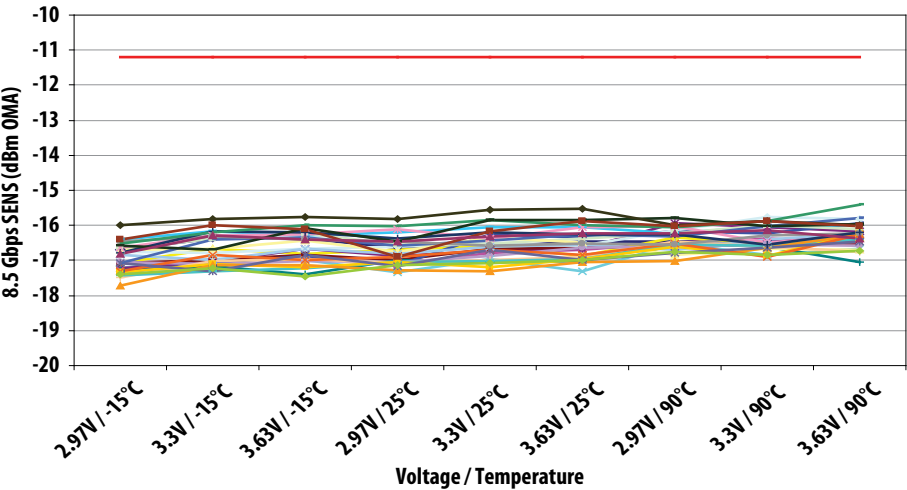


Figure 22. 8.5 Gbps SENS over the 9 test conditions.

Table 19. 4.25 Gbps SENS (dBm OMA), -12.1 dBm OMA max

	4.25 Gbps SENS								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-17.1	-16.9	-16.8	-17.4	-17.1	-17.0	-18.0	-17.9	-17.8
Min	-17.7	-17.3	-17.5	-17.9	-17.7	-17.7	-18.6	-18.6	-18.3
Max	-16.5	-16.1	-16.1	-16.7	-16.5	-16.3	-17.2	-16.9	-16.9

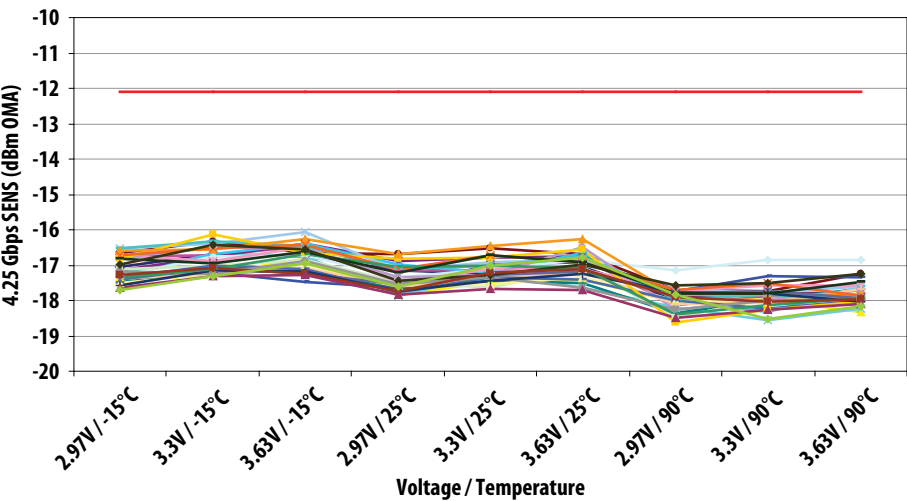


Figure 23. 4.25 Gbps SENS over the 9 test conditions.

Receiver Sensitivity (SENS)

Table 20. 2.125 Gbps SENS (dBm OMA), -13.1 dBm OMA max

	2.125 Gbps SENS								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-17.1	-16.8	-16.7	-17.4	-17.1	-17.0	-18.0	-17.8	-17.7
Min	-18.6	-18.4	-18.3	-18.5	-18.5	-18.3	-19.3	-19.0	-19.0
Max	-15.9	-15.0	-15.6	-16.0	-15.2	-16.0	-16.6	-16.7	-16.6

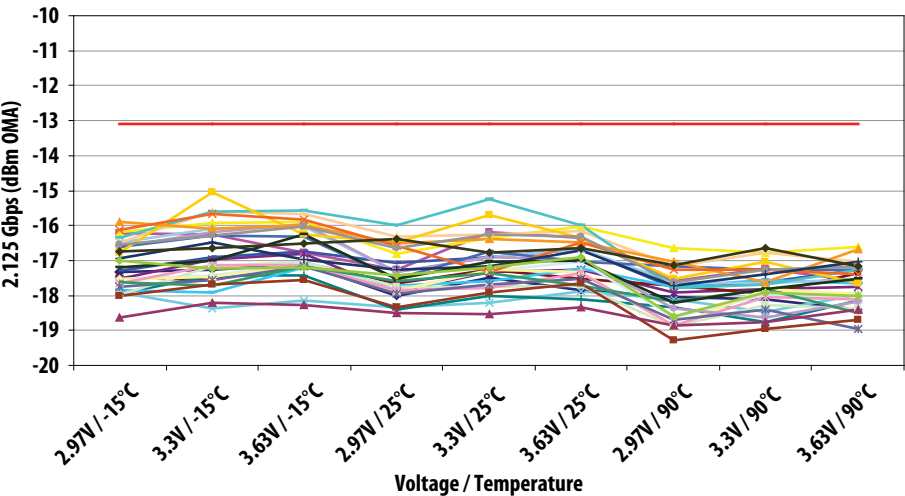


Figure 24. 2.125 Gbps SENS over the 9 test conditions.

8.5Gbps Stressed Receiver Sensitivity (SRS)

Table 21. 8.5 Gbps SRS (dBm OMA), -8.2 dBm OMA max

	8.5 Gbps SRS				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	-14.3	-14.2	-13.6	-13.1	-13.2
Min	-14.6	-14.7	-14.4	-13.7	-14.0
Max	-13.6	-13.3	-12.9	-12.7	-12.7

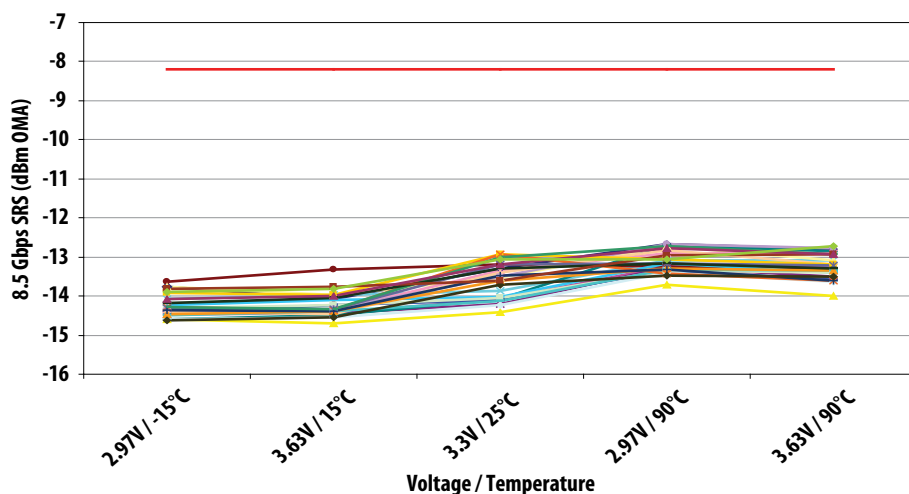


Figure 25. 8.5 Gbps SRS over the 5 test conditions.

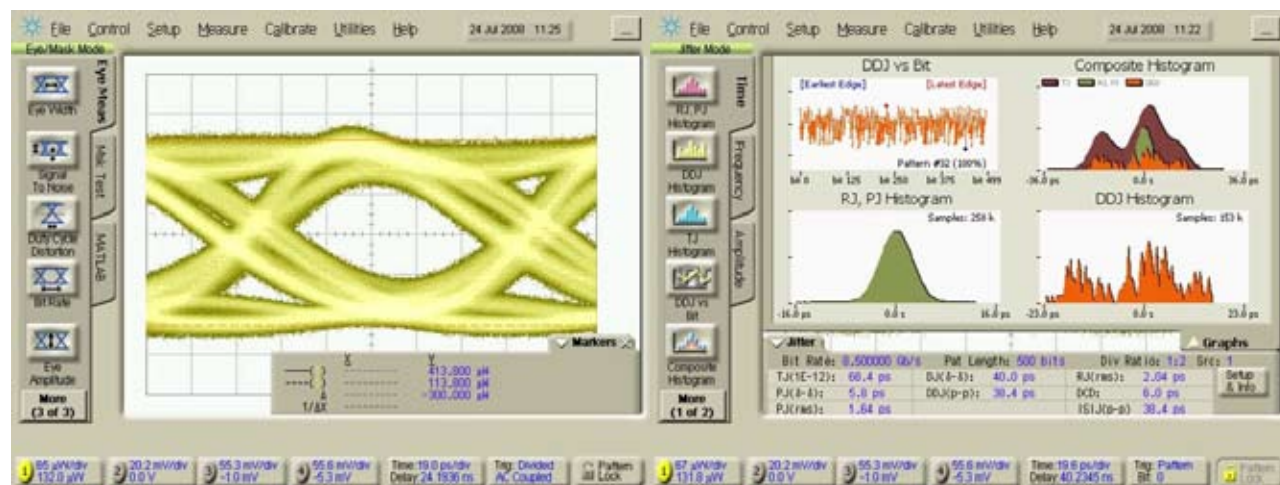


Figure 26. Optical input for 8.5Gbps stressed eye and jitter measurements.

2.125 Gbps Stressed Receiver Sensitivity (SRS)

Table 22. 2.125 Gbps SRS (dBm OMA), -10.8 dBm OMA max

	2.125 Gbps SRS				
	-15°C		25°C		90°C
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	-16.5	-16.3	-16.5	-16.5	-16.3
Min	-17.2	-17.0	-16.8	-16.9	-16.9
Max	-15.9	-15.6	-16.2	-16.0	-15.7

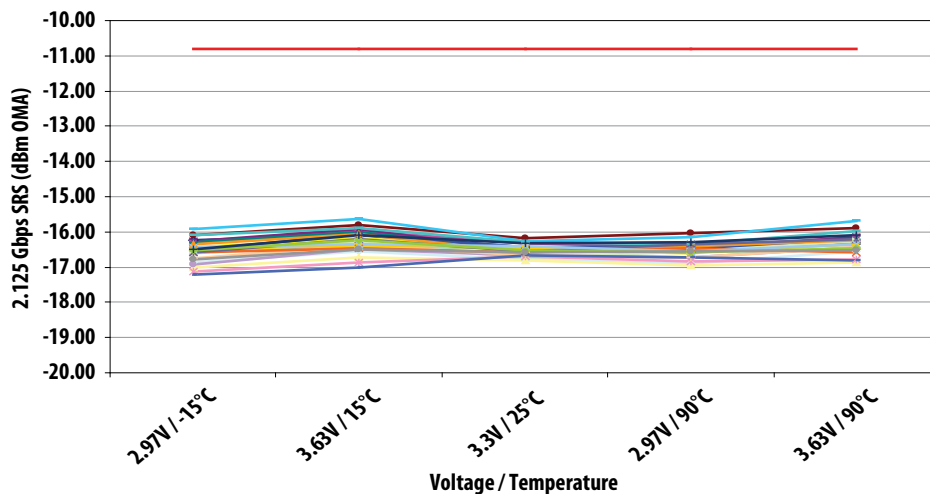


Figure 27. 2.125 Gbps SRS over the 5 test conditions.

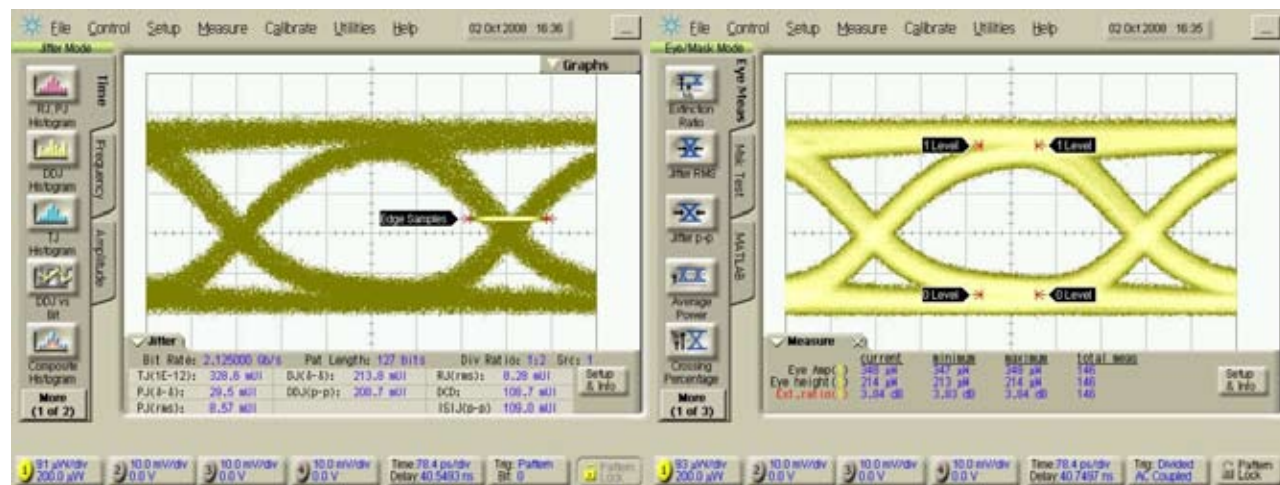


Figure 28. Input optical eye for 2.125 SRS measurements.

8.5Gbps Receiver Eye Mask Margin with Stressed Input (RX MM)

Table 23. 8.5 Gbps RX MM (%), 0% min

	8.5 Gbps RX MM				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	18	35	32	19	24
Min	13	29	27	15	19
Max	25	50	44	23	30

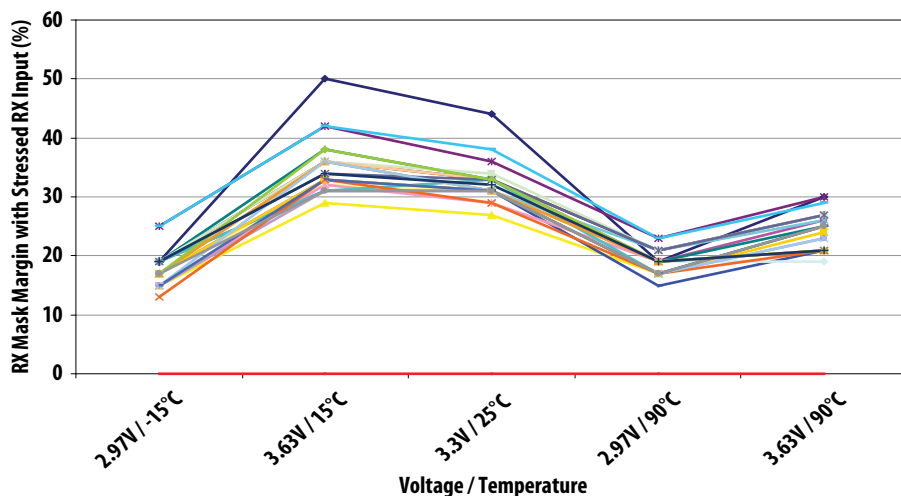


Figure 29. RX eye mask margin with stressed input over the 5 test conditions.

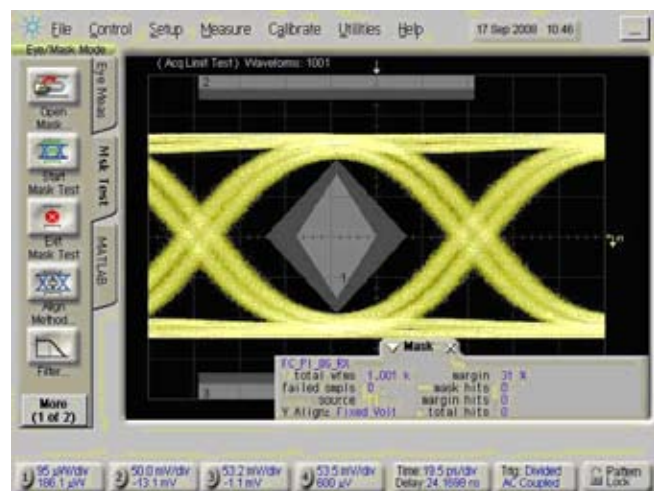


Figure 30. Typical RX eye mask with stressed input.

Receiver Differential Output Voltage Modulation Amplitude (RX VMA)

Table 24. RX VMA (mVpp), 340-850 mVpp

	RX VMA								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	458	488	497	465	498	508	481	515	528
Min	444	470	480	454	482	490	472	500	512
Max	481	528	539	479	528	536	494	529	542

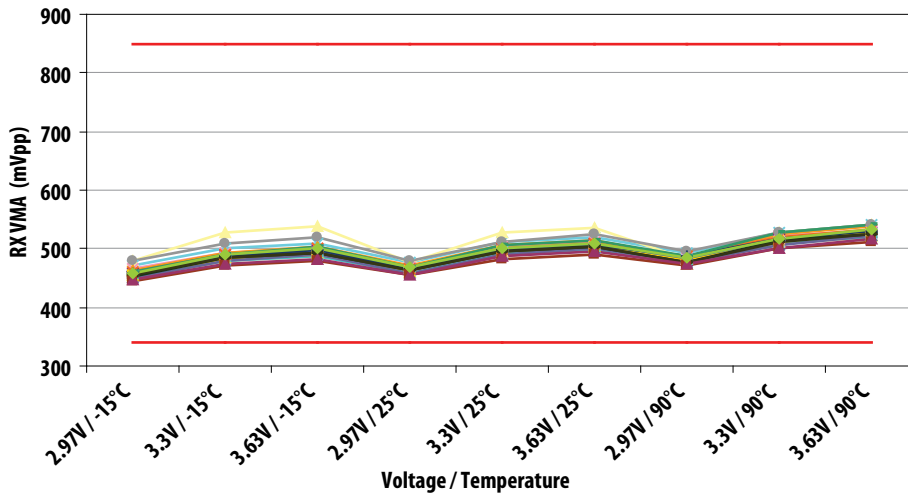


Figure 31. RX VMA over the 9 test conditions.

Receiver Output Total Jitter (RX DJ)

Table 25. RX TJ (UI), 0.71 UI max, Sample Size 10

	RX TJ				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	0.58	0.60	0.58	0.55	0.58
Min	0.53	0.55	0.53	0.49	0.55
Max	0.62	0.64	0.63	0.60	0.63

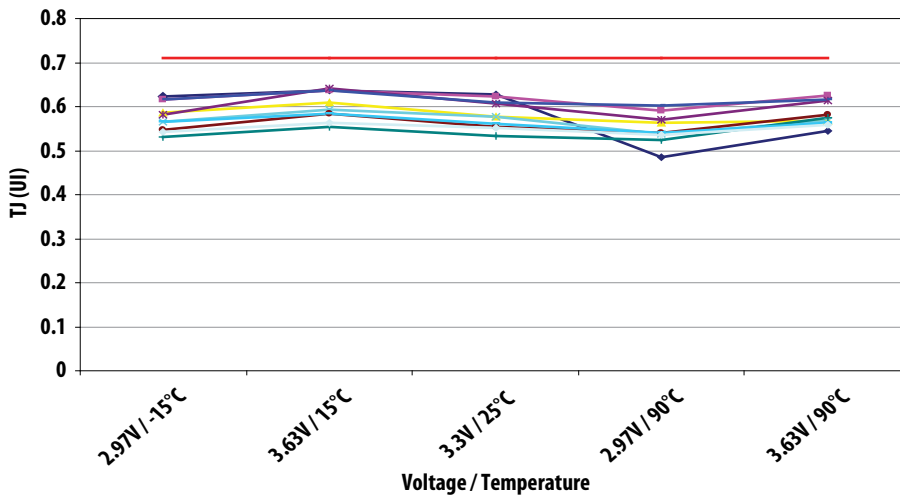


Figure 32. RX TJ over the 5 test conditions.

Receiver Output Deterministic Jitter (RX DJ)

Table 26. RX DJ (UI), 0.42 UI max, Sample Size 10

	RX DJ				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	0.26	0.28	0.26	0.24	0.26
Min	0.22	0.24	0.22	0.20	0.24
Max	0.32	0.32	0.30	0.30	0.30

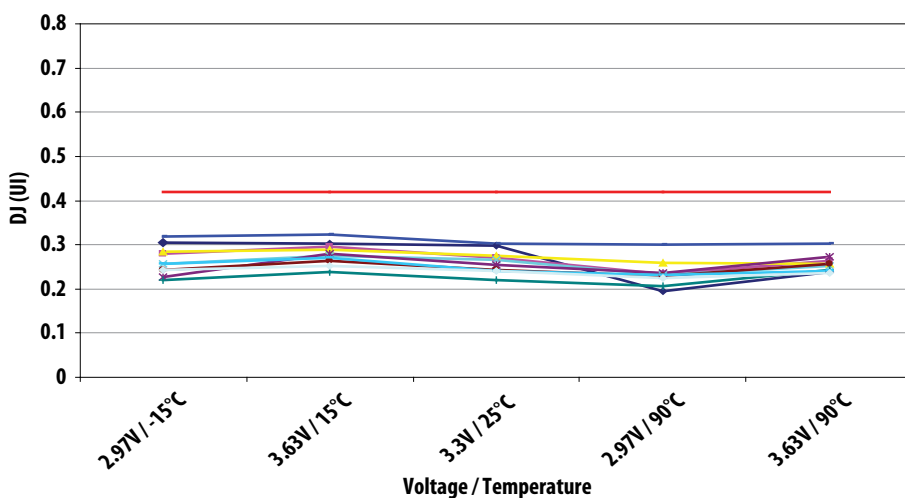


Figure 33. RX DJ over the 5 test conditions.

Receiver Output Data Dependent Pulse Width Shrinkage (RX DDPWS)

Table 27. RX DDPWS (UI), 0.36 UI max

	RX DDPWS				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	0.16	0.16	0.13	0.10	0.09
Min	0.12	0.12	0.08	0.07	0.06
Max	0.19	0.19	0.17	0.13	0.12

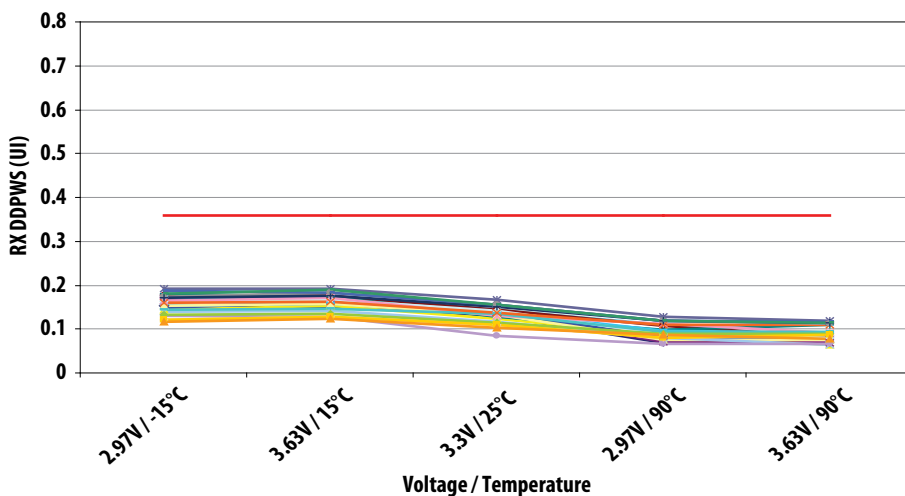


Figure 34. RX DDPWS over the 5 test conditions.

Receiver S-Parameters (SDD22, SCC22 and SCD22)

Spec per FC-PI-4 rev 8.0. 25/25 modules pass. Typical data are shown.

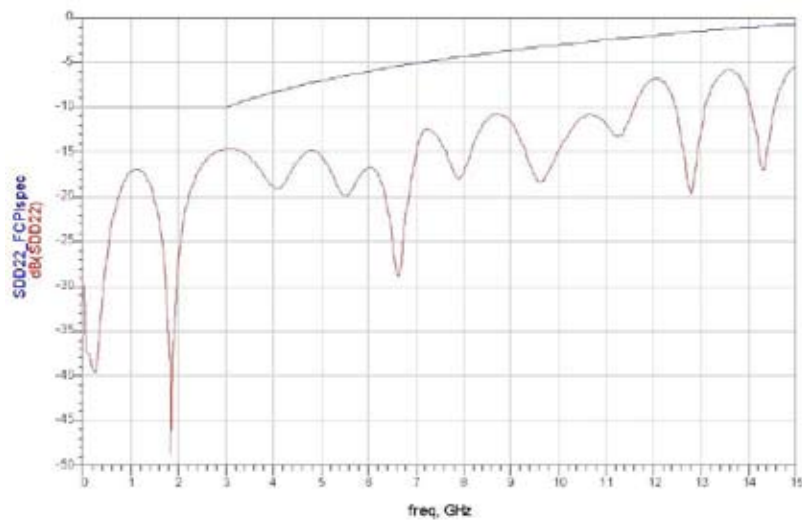


Figure 35. SDD22 at 25°C / 3.3V.

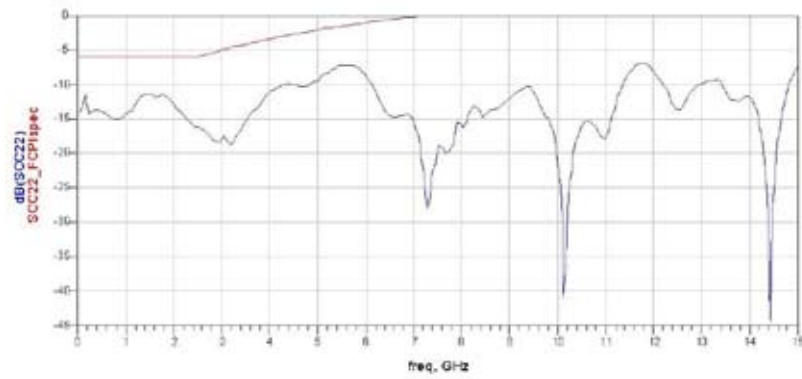


Figure 36. SCC22 at 25°C / 3.3V.

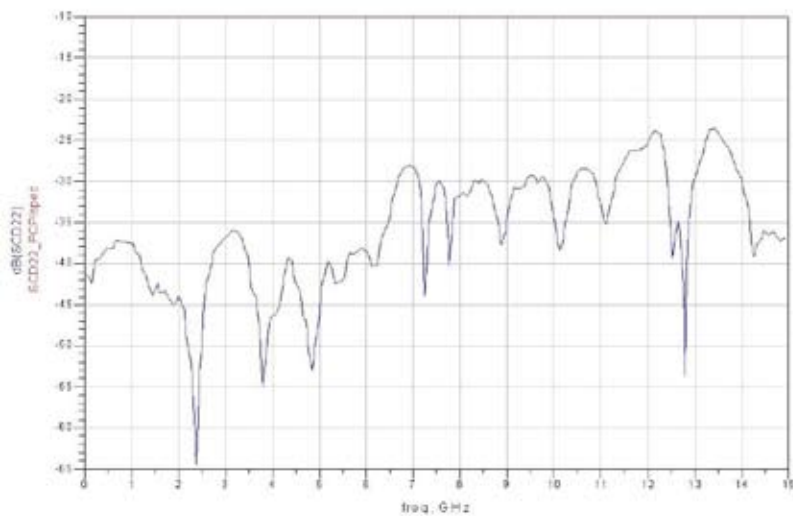


Figure 37. SCD22 at 25°C / 3.3V.

Receiver Common Mode Output Voltage (CMOV)

Table 28. CMOV (mV rms), 30 mV rms max, Sample Size 10

	CMOV				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	2.98	2.94	2.51	2.31	2.64
Min	2.30	2.14	2.09	2.01	2.06
Max	4.10	4.25	3.11	2.65	2.98

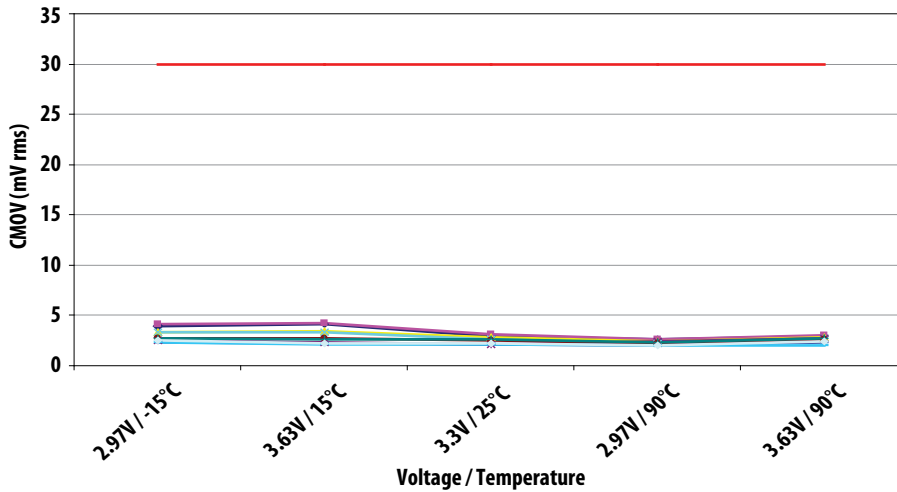


Figure 38. CMOV at the 5 test conditions.

Receiver Reflectance

Table 29. Receiver Reflectance (dB), 12 dB min, Sample Size 10

	Receiver Reflectance
	25°C, 3.3V
Ave	13.5
Min	13.6
Max	13.2

Receiver Maximum Average Input Power

Table 30. Receiver Maximum Average Input Power (dBm), +1 dBm min

	Receiver Maximum Average Input Power				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Input LOP with 0 errors in 300sec	>1.5	>1.5	>1.5	>1.5	>1.5

Loss of Signal Assert Level (LOSA)

Table 31. LOSA (dBm OMA), -15.6 dBm OMA max

	LOSA								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-19.9	-20.1	-20.1	-19.3	-19.4	-19.5	-18.2	-18.4	-18.6
Min	-20.5	-20.7	-20.7	-19.8	-20.1	-18.4	-18.8	-19.0	-19.1
Max	-19.4	-19.5	-19.5	-18.5	-18.5	-20.9	-17.7	-17.8	-18.0

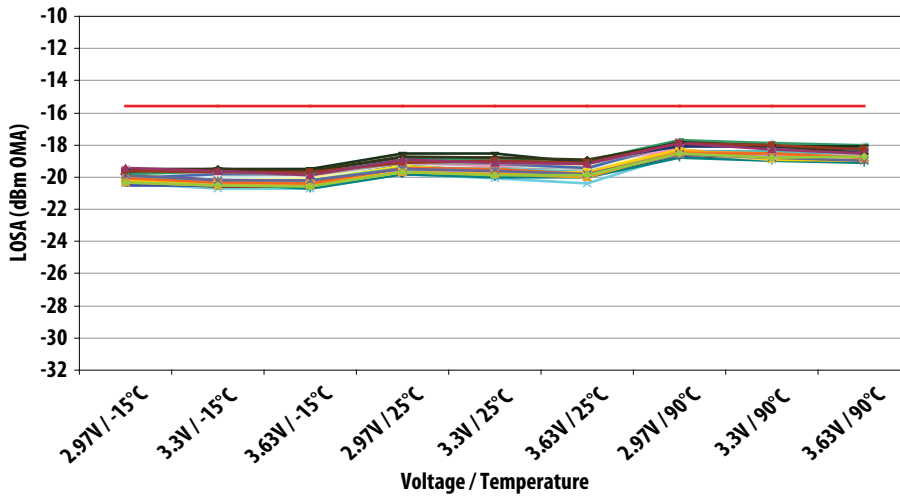


Figure 39. LOSA over the 9 test conditions.

Loss of Signal De-assert Level (LOSD)

Table 32. LOSD (dBm OMA), -13.1 dBm OMA min

	LOSD								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-17.6	-17.6	-17.6	-16.9	-17.0	-17.1	-15.9	-16.1	-16.2
Min	-18.2	-18.3	-18.3	-17.7	-17.8	-17.9	-16.7	-16.7	-16.8
Max	-17.0	-16.9	-16.9	-16.1	-16.2	-16.5	-15.4	-15.5	-15.6

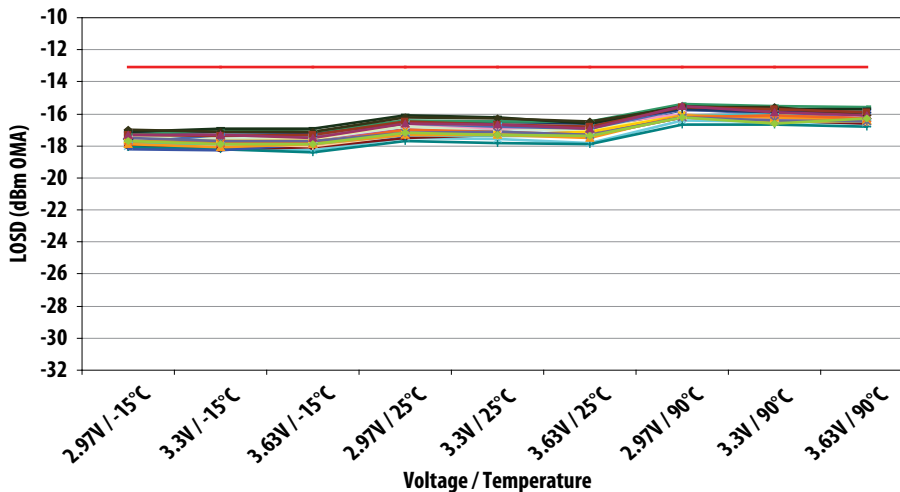


Figure 40. LOSD over the 9 test conditions.

Loss of Signal Hysteresis (LOSH)

Table 33. LOSH (dB), 0.5 dB min

	LOSH								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	2.4	2.4	2.5	2.4	2.4	2.4	2.3	2.3	2.4
Min	2.1	2.2	2.3	2.1	2.1	2.1	2.1	2.2	2.1
Max	2.7	2.6	2.8	2.6	2.6	2.6	2.5	2.5	2.5

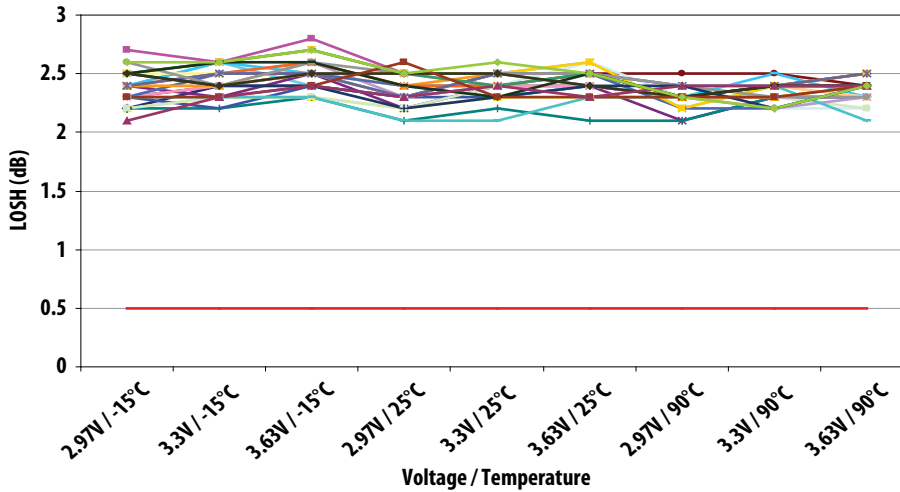


Figure 41. LOSH over the 9 test conditions.

Diagnostic Monitoring Interface Voltage Accuracy (DMI VCC)

Table 34. DMI VCC (V), +/- 0.1 V max

	DMI VCC								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.01	0.01	-0.03	0.00	0.00	-0.03	0.00	0.00	-0.03
Min	-0.02	-0.02	-0.05	-0.03	-0.02	-0.05	-0.02	-0.02	-0.05
Max	0.04	0.02	0.01	0.02	0.03	-0.01	0.02	0.03	-0.01

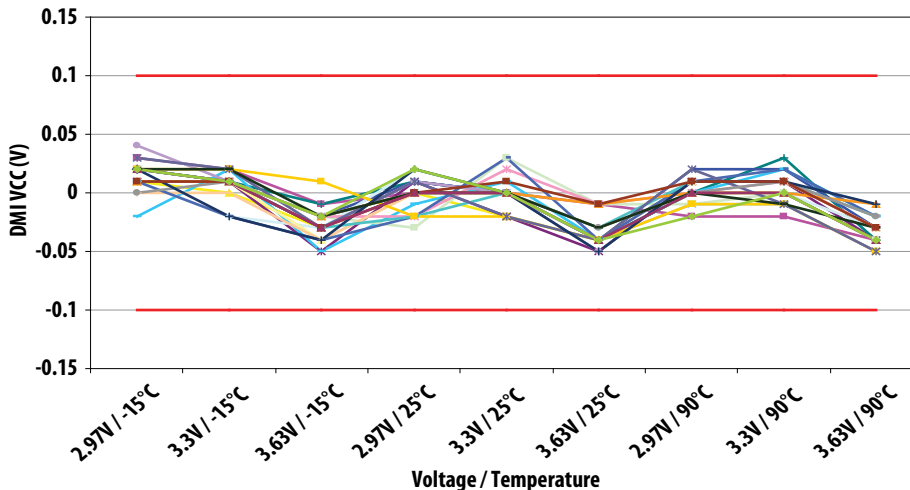


Figure 42. DMI VCC over the 9 test conditions.

Diagnostic Monitoring Interface Voltage Accuracy (DMI TEMP)

Table 35. DMI TEMP (C), +/- 3C max

	DMI TEMP								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.1	0.2	0.7	-0.9	-0.2	0.4	0.1	1.0	1.6
Min	-1.7	-0.9	-0.7	-1.9	-0.9	-0.9	-1.0	0.1	0.1
Max	1.3	1.3	1.9	-0.3	0.9	2.0	1.1	2.0	2.4

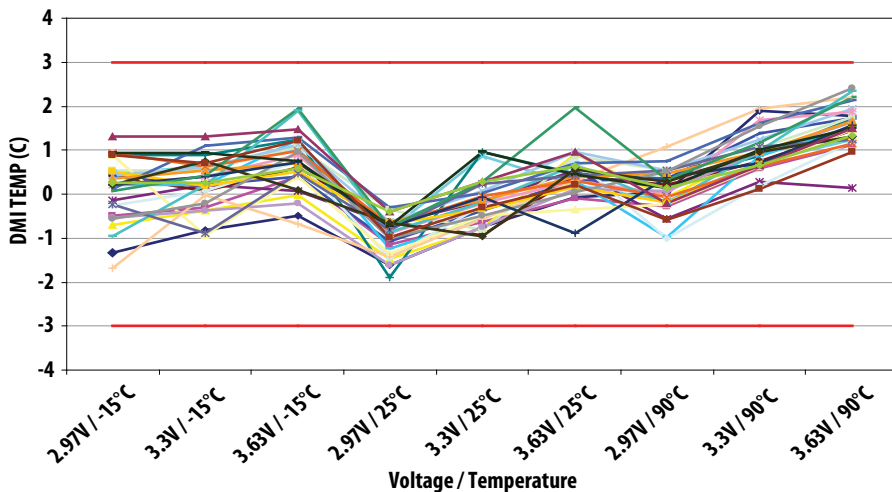


Figure 43. DMI TEMP over the 9 test conditions.

Diagnostic Monitoring Interface LOP Accuracy (DMI LOP)

Table 36. DMI LOP (dB), +/- 3 dB max

	DMI LOP								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	-0.03	-0.03	0.00	-0.11	-0.07	0.00	-0.09	-0.05	0.01
Min	-0.13	-0.17	-0.13	-0.25	-0.20	-0.15	-0.20	-0.14	-0.11
Max	0.17	0.17	0.11	0.17	0.03	0.07	0.01	0.05	0.11

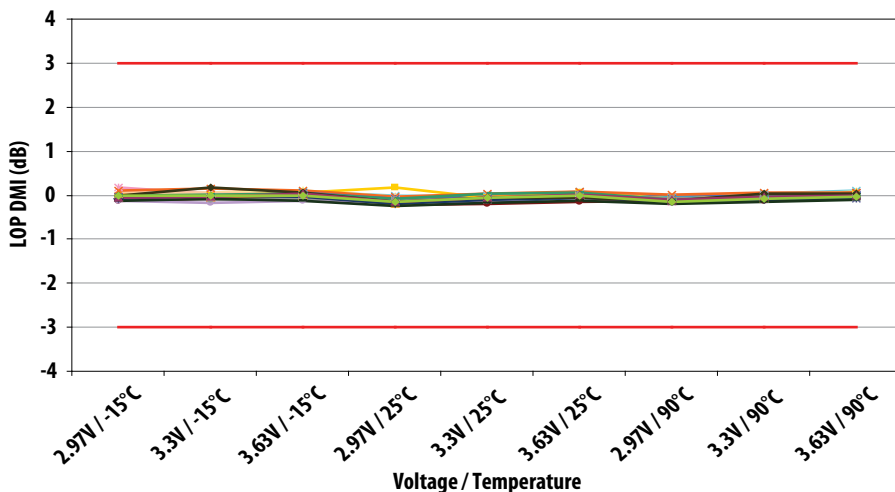


Figure 44. DMI LOP over the 9 test conditions.

Diagnostic Monitoring Interface RX Accuracy (DMI RX)

Table 37. DMI RX with -3 dBm Input (dB), +/- 3 dB max

	DMI RX								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.23	0.18	0.18	0.27	0.24	0.24	0.41	0.37	0.37
Min	-0.41	-0.65	-0.70	-0.71	-0.73	-0.48	-0.22	-0.23	-0.35
Max	0.81	0.64	0.72	0.71	0.70	0.63	0.83	0.83	0.86

Table 38. DMI RX with -6 dBm Input (dB), +/- 3 dB max

	DMI RX								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.39	0.37	0.31	0.46	0.41	0.37	0.57	0.53	0.54
Min	-0.26	-0.53	-0.53	-0.60	-0.63	-0.38	-0.14	-0.17	-0.27
Max	0.96	0.87	0.81	1.21	0.84	0.78	1.22	0.94	1.01

Table 39. DMI RX with -9 dBm Input (dB), +/- 3 dB max

	DMI RX								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.45	0.37	0.34	0.48	0.37	0.37	0.56	0.52	0.51
Min	-0.23	-0.55	-0.50	-0.59	-0.69	-0.34	-0.11	-0.11	-0.87
Max	1.04	0.82	0.85	1.52	0.85	0.79	0.97	0.93	1.04

Diagnostic Monitoring Interface RX Accuracy (DMI RX)

Table 40. DMI RX with -12 dBm Input (dB), +/- 3 dB max

	DMI RX								
	-15°C			25°C			90°C		
	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V	2.97V	3.30V	3.63V
Ave	0.48	0.37	0.27	0.37	0.42	0.36	0.43	0.49	0.52
Min	-0.88	-0.61	-0.87	-0.88	-0.71	-0.46	-0.89	-0.82	-0.34
Max	1.04	0.88	0.80	1.01	0.92	0.83	0.96	0.97	0.96

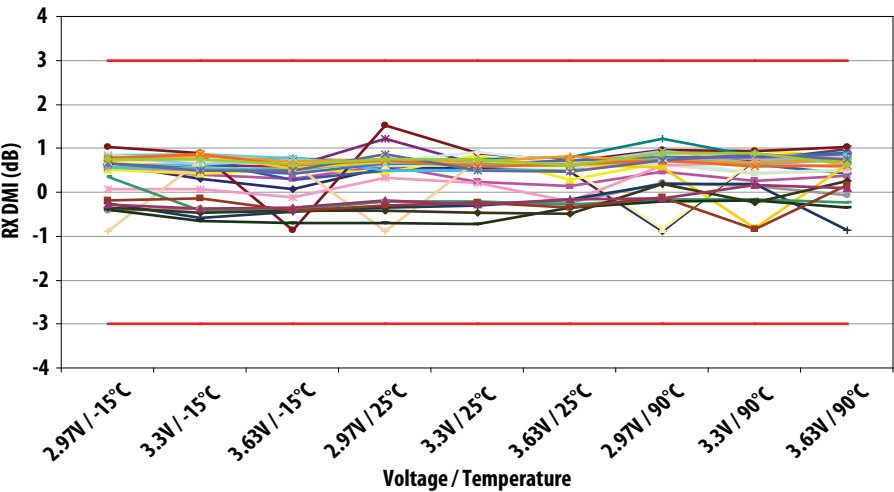


Figure 45. Worst case DMI RX over the 9 test conditions and 4 input powers.

Inrush Current During Hotplug

Sample Size 5



Figure 46. Typical TX inrush current during hotplug (62.5 mA per division).

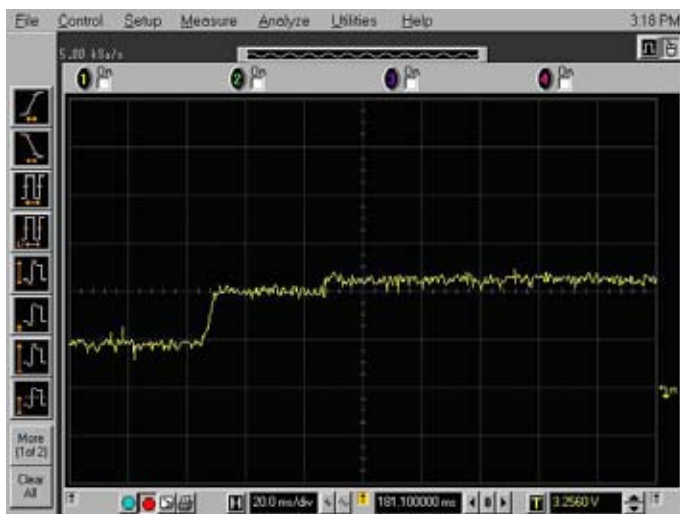


Figure 47. Typical RX inrush current during hotplug (62.5 mA per division).

LOS Assert Timing

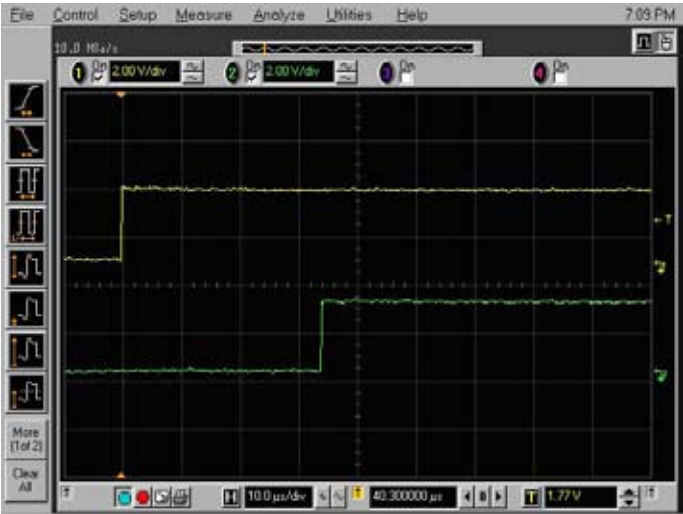


Figure 48. Typical LOS assert timing. Yellow trace: inverse of input power to module, green trace: LOS.

Table 41. LOS Assert Timing (us), 100 us max, Sample Size 5

	LOS Assert Timing				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	36.4	37.8	35.6	31.0	33.0
Min	34.0	36.0	34.0	28.0	30.0
Max	39.0	40.0	39.0	34.0	36.0

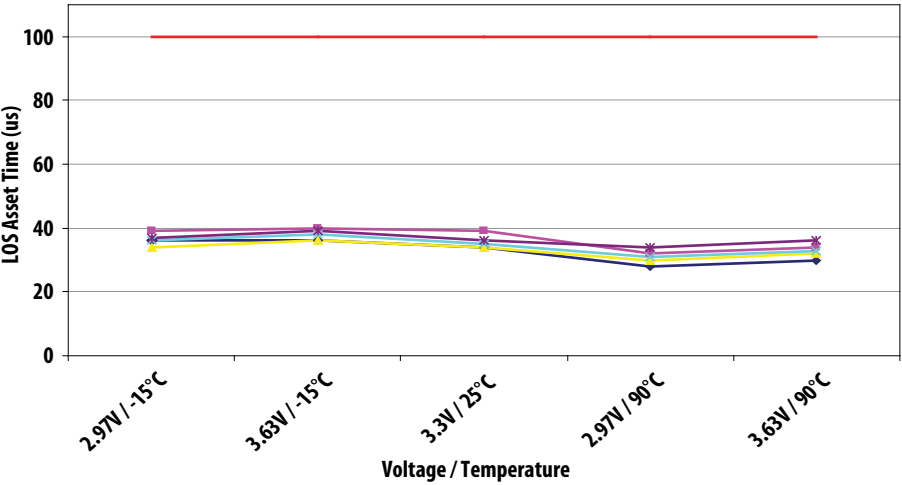


Figure 49. LOS Assert timing over the 5 test conditions.

LOS De-assert Timing



Figure 50. Typical LOS de-assert timing. Yellow trace: inverse of input power to module, green trace: LOS.

Table 42. LOS De-assert Timing (us), 100 us max, Sample Size 5

	LOS De-assert Timing				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Ave	81.0	80.0	79.4	49.8	51.4
Min	79.0	78.0	77.0	48.0	50.0
Max	83.0	82.0	81.0	54.0	54.0

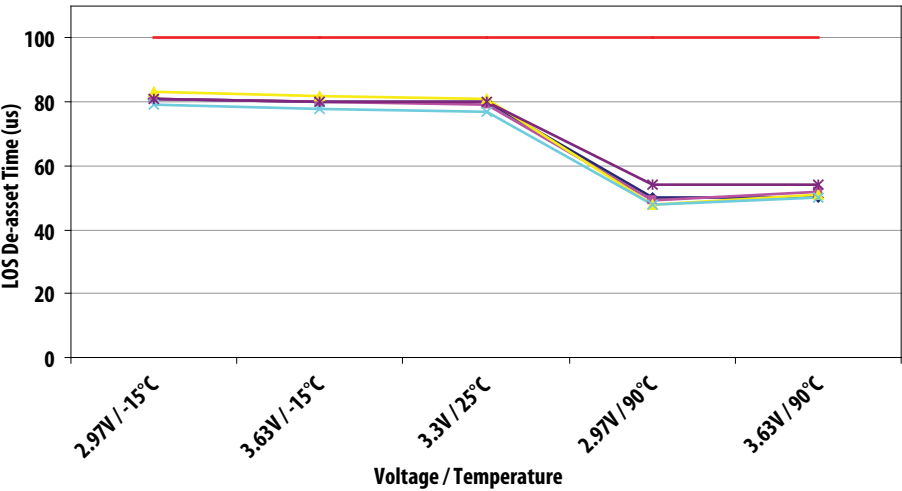


Figure 51. LOS De-assert Timing at the 5 test conditions.

TX_DISABLE Assert Timing

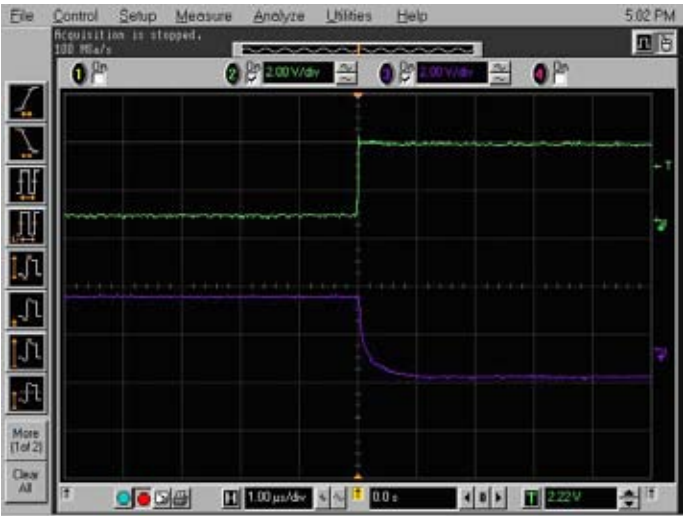


Figure 52. Typical TX_DISABLE Assert timing. Green trace: TX_DISABLE, purple trace: LOP.

Table 43. TX DISABLE Assert Timing (us), 10 us max, Sample Size 5

	TX_DISABLE Assert Timing				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Result	<1	<1	<1	<1	<1

TX_DISABLE De-assert Timing

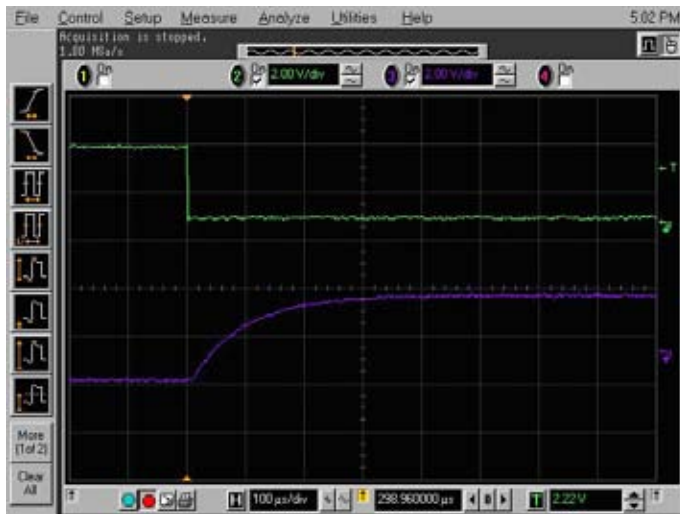


Figure 53. Typical TX_DISABLE de-assert timing. Green trace: TX_DISABLE, purple trace: LOP.

Table 44. TX_DISABLE De-assert Timing (ms), 1 ms max, Sample Size 5

	TX_DISABLE De-assert Timing				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63
Result	<0.35	<0.35	<0.35	<0.35	<0.35

Module Initialization Timing

Table 45. Module Initialization Timing (ms), 300 ms max, Sample Size 5

	Receiver Reflectance
	25°C, 3.3V
Result	<175



Figure 54. Module initialization time at 25C and 3.3V. Yellow trace: VCC, purple trace: LOP.

PSNR

Table 46. Transmitter Optical Mask Margin Penalty from Supply Noise (%), Sample Size 5

	Mask Margin Penalty from Supply Noise				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63V
Ave	10	11	11	8	9
Min	7	8	9	4	5
Max	12	17	14	10	15

Table 47. Receiver Sensitivity Penalty from Supply Noise (dB), Sample Size 5

	Mask Margin Penalty from Supply Noise				
	-15°C		25°C	90°C	
	2.97V	3.63V	3.3V	2.97V	3.63V
Ave	0.22	0.53	0.36	0.28	1.00
Min	0.01	0.21	0.27	0.12	0.43
Max	0.77	0.71	0.50	0.58	1.38

Live Traffic ESD

Table 48. ESD Errors (packets with errors / second), 1% max, Sample Size 10

	Contact	Air
Worst Case	0.00013%	0.00010%

EMI

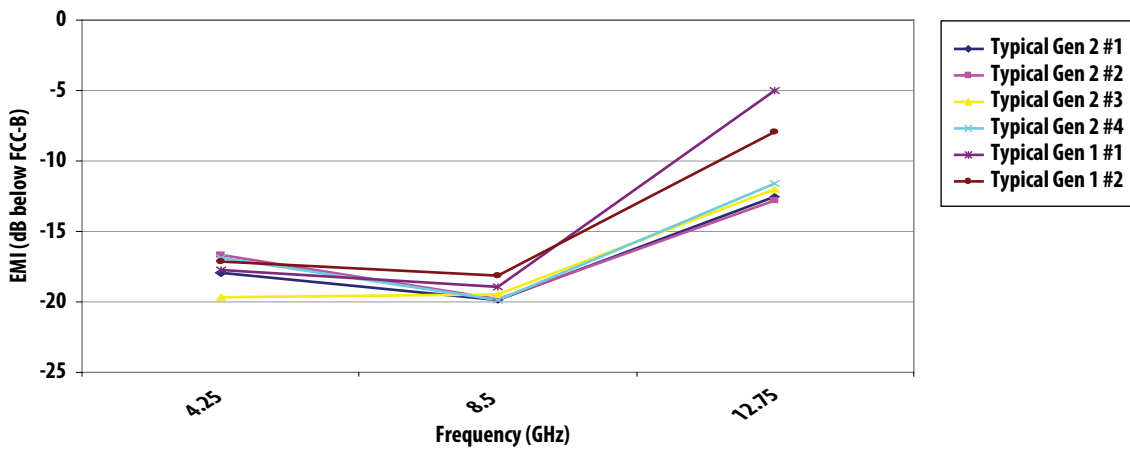


Figure 55. Typical EMI in GTEM with 1010 pattern at 4.25 GHz. Worst case EMI is >5 dB better than previous 8G module design.

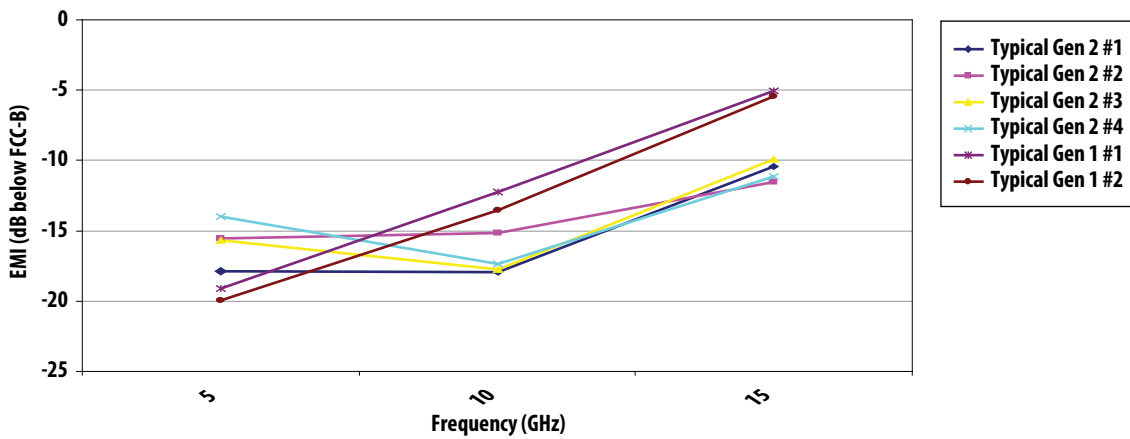


Figure 56. Typical EMI in GTEM with 1010 pattern at 5 GHz. Worst case EMI is >4 dB better than previous 8G module design.

Radiated Immunity

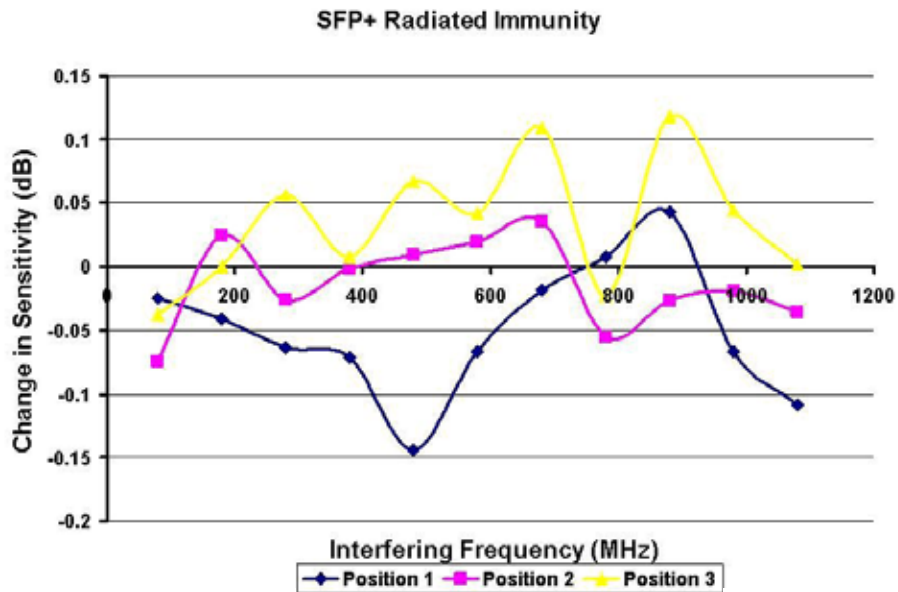


Figure 57. Sensitivity penalty due to incident EM fields at 3 orthogonal positions is less than 0.15 dB.

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